Classification of quantum graphs over complex 3×3 matrices

(Klasyfikacja grafów kwantowych nad zespolonymi macierzami 3×3)

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Master's thesis

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Abstract

According to all known laws of aviation, there is no way a bee should be able to fly. Its wings are too small to get its fat little body off the ground. The bee, of course, flies anyway because bees don't care what humans think is impossible.

Według wszystkich znanych praw lotnictwa, pszczoła nie powinna w ogóle latać. Jej skrzydła są zbyt małe, by unieść jej pulchne ciałko z ziemi. Pszczoła, oczywiście, lata mimo wszystko, ponieważ pszczoły nie przejmują się tym, co ludzie uważają za niemożliwe.

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Chapter 1

Introduction

1.1 Motivation

Quantum graphs is a topic that has seen a resurgence in interest in recent years.

1.2 Scope

This thesis is organised as follows.

We begin, in the preliminary chapter, by recalling the algebraic background required for everything that follows: dual groups and self-duality of finite abelian groups, positive and faithful functionals on C^*-algebras, group actions and crossed products, and the theory of matrix Lie groups and Lie algebras – in particular the unitary groups $U(n)$, $SU(n)$, and $PU(n)$, together with their associated Lie algebras $\mathfrak{gl}_n$, $\mathfrak{sl}_n$, and $\mathfrak{su}_n$, which will serve throughout as the spaces in which quantum graphs are classified.

Building on this foundation, we then introduce the central objects of the thesis: *quantum sets* and *quantum graphs*, following Schäfer and Tobolski [9]. We show how classical sets and matrix algebras both give rise to quantum sets, define quantum graphs as Schur-idempotent, $*$-preserving maps generalising classical adjacency matrices, and extend the classical theory of voltage and derived graphs to this setting. A key result shows that derived quantum graphs built from different group actions on the same voltage data are always quantum isomorphic, a tool that will later let us relate quantum graphs on M_3 back to classical graphs. We close the chapter by recalling the notion of *quantum isomorphism*, and by stating Gromada's theorem, which translates quantum graphs on M_n into an explicit correspondence with linear subspaces of $\mathfrak{gl}_n$.

The remainder of the thesis is devoted to quantum graphs over 3×3 matrices. Using the generalised Pauli (clock-and-shift) matrices P, Q, we construct and completely classify, up to unitary equivalence, all loop-free, undirected quantum graphs on M_3 built from this family, showing that they fall into exactly five isomorphism classes. We then realise each nontrivial class concretely as the derived quantum graph of an explicit classical voltage graph over $\mathbb{Z}_3$, and show that every such graph is vertex-transitive. Finally, we show that the clean, single-invariant classification available for M_2 – resting on the exceptional isomorphism $PU(2) \cong SO(3)$ – provably fails to generalise to M_3, where infinite families of mutually non-isomorphic quantum graphs exist for each fixed number of quantum edges, and where quantum graphs can be exhibited that are not quantum isomorphic to any classical graph at all.

1.3 Acknowledgements

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Chapter 2

Preliminaries

The following sections cover various topics in algebra. They are not intended to provide an exhaustive treatment; rather, they are meant to equip the Reader with the necessary background to understand the remainder of this work. References to further information are provided where appropriate.

2.1 Group theory

In the following section we will introduce some group theoretic notions that be used throughout this thesis. We assume that the Reader is familiar with the notion of a group. By $\mathbb{T}$ we will denote the torus, i.e. the set of complex numbers with the absolute value of 1, along with multiplication as its group action. By e_G we will denote the identity element of G. If G is abelian, its identity element will be expressed by 1_G.

Definition 2.1.1 (Dual of a group)**.** Let G be a group. The *dual group* $\hat{G}$ is the set of all homomorphisms from G into the torus $\mathbb{T}$ with the group operation defined as

$$(\varphi\psi)(g) = \varphi(g)\psi(g)$$

for all $\varphi, \psi \in \hat{G}$ and $g \in G$, where the latter multiplication takes place in $\mathbb{C}$.

Remark 2.1.1. We will often call the elements of the dual group *characters*. If a group is isomorphic to its complement it will be called *self-dual*.

The following proposition will be used throughout the thesis, as almost all of the groups we will be dealing with will be finite and abelian.

Proposition 2.1.1. *If G is finite and abelian, then it is self-dual.*

Proof. First we will prove the statement for finite cyclic groups. Let $G = \langle g \rangle$, $\mathrm{ord}(g) = n$. We will prove that the dual is cyclic of order n. Let $\chi \colon G \to \mathbb{T}$

be a character. Since the group is cyclic, the character is entirely determined by the value it takes on the generator g. Since $g^n = e$, obviously $\chi(g)^n = 1$, so $\chi(g) \in \mu_n = \{\exp(2k\pi i/n) : k = 0, 1, \ldots, n-1\} \subseteq \mathbb{T}$. For each possible choice of $\chi(g) \in \mu_n$ we get a different character. It is then easy to see that $\chi : G \to \mathbb{T}$ such that $\chi(g) = \exp(2\pi i/n)$ generates this group, i.e. $\hat{G} = \{\chi, \chi^2, \ldots, \chi^{n-1}, \chi^n = e_{\hat{G}}\}$.

Now since every finite abelian group can be written as a finite product of finite cycles, all we need to do is check whether the statement holds for a product of two groups A, B that are themselves self-dual, i.e. $\widehat{A \times B} \cong \widehat{A} \times \widehat{B}$. This is self-evident as every character $\chi \in \widehat{A \times B}$ is determined by its restrictions to $A \times \{e_B\}$ and $\{e_A\} \times B$, since

$$\chi(a, b) = \chi((a, e_B)(e_A, b)) = \chi(a, e_B)\chi(e_A, b).$$

Thus $\chi \leftrightarrow (\chi|_A, \chi|_B)$ is an isomorphism. □

2.2 C^*-algebras

In the following section we will introduce some notions stemming from the theory of C^*-algebras. We assume the Reader is familiar with the notion of a C^*-algebra. For reference, see Blackadar [2].

Multiplication in a C^*-algebra A is a bilinear map

$$m \colon A \times A \to A.$$

By the universal property of the tensor product, m extends uniquely to a linear map

$$m^* \colon A \otimes A \to A.$$

Throughout this thesis, we will use the same notation (m) for both the bilinear multiplication and its unique linear extension, relying on the context to distinguish between the two.

Definition 2.2.1 (Linear functional)**.** A *linear functional* on a C^*-algebra A is a linear function $\psi : A \to \mathbb{C}$.

Remark 2.2.1. We will call linear functionals simply as *functionals.*

The following notions will be useful for defining an inner product on C^*-algebras. Positiveness and faithfulness are all we need to define a norm on a finite-dimensional C^*-algebra equipped with a suitable functional.

Definition 2.2.2 (Positive functional)**.** A functional ψ on a C^*-algebra A is called *positive*, if

$$0 \leq \psi(a^*a)$$

for any $a \in A$.

Definition 2.2.3 (Faithful functional)**.** A positive functional ψ on a C^*-algebra A is called *faithful*, if

$$\psi(a^*a) = 0 \Leftrightarrow a = 0$$

for any $a \in A$.

Remark 2.2.2. In the literature, the functional ψ is often called a *state*.

Example 2.2.1 (Positive and faithful functionals on matrix algebras)*.* Let $A = M_n(\mathbb{C})$, the C^*-algebra of $n \times n$ complex matrices, equipped with the usual trace Tr. Define

$$\psi(a) = n \operatorname{Tr}(a), \qquad a \in A.$$

Since a^*a is positive semi-definite for every $a \in A$, its eigenvalues are non-negative, and hence $\operatorname{Tr}(a^*a) \geq 0$; thus ψ is a positive functional. Moreover, $\operatorname{Tr}(a^*a) = 0$ forces all eigenvalues of a^*a to vanish, which (since a^*a is positive semi-definite, hence diagonalizable with non-negative eigenvalues) implies $a^*a = 0$ and therefore $a = 0$. So ψ is faithful.

Example 2.2.2 (A positive functional that is not faithful)*.* Let $A = \mathbb{C}^2 = \mathbb{C} \oplus \mathbb{C}$, with coordinatewise multiplication and involution given by complex conjugation in each coordinate. Define

$$\psi(x, y) = x, \qquad (x, y) \in A.$$

For $(x, y) \in A$ we have $(x, y)^*(x, y) = (|x|^2, |y|^2)$, so $\psi\big((x, y)^*(x, y)\big) = |x|^2 \geq 0$, and ψ is positive. However $\psi\big((0, y)^*(0, y)\big) = 0$ for every $y \in \mathbb{C}$, while $(0, y) \neq 0$ whenever $y \neq 0$. Hence ψ is positive but *not* faithful.

The following notion is akin to the notion of a group action on a set.

Definition 2.2.4 (Group action on a C^*-algebra)**.** By *group action* of a group G on a C^*-algebra A we mean a group homomorphism

$$\alpha\colon G \to \operatorname{Aut}(A).$$

We will write $g \cdot a$ instead of $\alpha(g)(a)$ then the action in question is clear.

Example 2.2.3 (Group actions on C^*-algebras)*.*

1. *Conjugation action.* Let $A = M_n(\mathbb{C})$ and let $G = U(n)$ be the group of unitary matrices. Then G acts on A via

$$\alpha(u)(a) = uau^*, \qquad u \in U(n),\ a \in A.$$

   Each $\alpha(u)$ is easily checked to be a $*$-automorphism of A, and $\alpha(u_1u_2) = \alpha(u_1)\alpha(u_2)$, so $\alpha\colon G \to \operatorname{Aut}(A)$ is indeed a group homomorphism.

2. *Translation action.* Let $A = C(\mathbb{T})$, the continuous complex-valued functions on the torus $\mathbb{T}$, and let $G = \mathbb{Z}$ act by rotation:

$$(n \cdot f)(z) = f(z - n\theta \bmod 2\pi), \qquad n \in \mathbb{Z},\ f \in A,$$

for a fixed $\theta \in \mathbb{R}$. This is precisely the action used to construct the irrational rotation algebra as a crossed product (see Example 2.2.5).

3. *Trivial action.* For *any* C^*-algebra A and *any* group G, setting $\mathrm{triv}(g)(a) = a$ for all $g \in G,\ a \in A$ trivially defines a group action, since $\mathrm{triv}(g) = \mathrm{id}_A \in \mathrm{Aut}(A)$ for every g, and the identity map is trivially a group homomorphism from G into $\mathrm{Aut}(A)$.

2.2.1 Crossed-product C^*-algebra

In this subsection we introduce the notion of a crossed-product C^*-algebra, in the limited form needed for our purposes; for a more general treatment we refer the Reader to Blackadar [2]. The definitions below follow Sierakowski [10].

Definition 2.2.5 (C^*-dynamical system)**.** A *C^*-dynamical system* is a triple (A, G, α), where A is a C^*-algebra, G is a discrete group, and $\alpha\colon G \to \mathrm{Aut}(A)$ is a group action on A. For a C^*-dynamical system (A, G, α) we set

$$C_c(G, A, \alpha) = \{f\colon G \to A : f \text{ has finite support}\},$$

the space of all A-valued functions on G. We usually write $g \cdot a$ for $\alpha(g)(a)$.

If G is finite, every $f : G \to A$ is automatically finitely supported. A typical element $a \in C_c(G, A, \alpha)$ is written $a = \sum_{g\in G} a_g u_g$, where only finitely many of the $a_g \in A$ are nonzero, and $\{u_g\}_{g\in G}$ is a formal family of symbols indexed by G (which specific representation the u_g correspond to depends on the crossed product in question).

The space $C_c(G, A, \alpha)$ carries both a product and an involution, defined precisely so that the group action becomes *inner*, i.e. so that $g \cdot a = u_g a u_g^*$ for all $g \in G$, $a \in A$. Explicitly, for $a = \sum_{g\in G} a_g u_g$ and $b = \sum_{h\in G} b_h u_h$ in $C_c(G, A, \alpha)$, the product and involution are given by

$$ab = \sum_{g,h\in G} a_g\,(g \cdot b_h)\,u_{gh}, \qquad a^* = \sum_{g\in G} (g^{-1} \cdot a_g^*)\,u_{g^{-1}}.$$

Definition 2.2.6 (Covariant representation)**.** Fix a dynamical system (A, G, α). A *covariant representation* $(\pi, u, \mathcal{H})$ of (A, G, α) consists of a unitary representation $u\colon G \to B(\mathcal{H})$ of G and a representation $\pi\colon A \to B(\mathcal{H})$ of A such that

$$u(g)\,\pi(a)\,u(g)^* = \pi(g \cdot a)$$

for every $g \in G$ and $a \in A$.

Definition 2.2.7 (Full C^*-algebra norm on $C_c(G, A, \alpha)$)**.** The *full C^*-algebra norm* on $C_c(G, A, \alpha)$ is given by

$$\|\cdot\| = \sup \|(\pi \times u)(\cdot)\|,$$

where the supremum ranges over all covariant representations $(\pi, u, \mathcal{H})$ of (A, G, α).

Definition 2.2.8 (Crossed product C^*-algebra)**.** The *crossed product* of a C^*-algebra A and a discrete group G is the completion of $C_c(G, A, \alpha)$ with respect to the full norm. We denote it by

$$\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}.$$

Remark 2.2.3. In this work we will not make use of norms of crossed product C^*-algebras and almost all groups will be discrete, so we can treat crossed product C^*-algebras as formal linear combinations of some family $\{u_g\}_{g \in G}$, which is dependent on the group action in question.

The Reader will find some examples of such C^*-algebras below (Examples 2.2.4 and 2.2.5).

Example 2.2.4 (Trivial action recovers the group C^*-algebra). Let $A = \mathbb{C}$ with the trivial action $\alpha_g = \mathrm{id}$ for all $g \in G$. Then every element of $C_c(G, A, \alpha)$ is just a finitely supported function $G \to \mathbb{C}$, with convolution product and involution reducing to the usual group algebra structure, since the action contributes no twisting. Completing in the full norm yields

$$\mathbb{C} \rtimes_\alpha G \;\cong\; C^*(G),$$

the (full) group C^*-algebra of G. For instance, taking $G = \mathbb{Z}$ gives $\mathbb{C} \rtimes \mathbb{Z} \cong C(\mathbb{T})$, since $C^*(\mathbb{Z})$ is generated by a single unitary with no relations, i.e. the continuous functions on the circle via the Fourier transform.

Example 2.2.5 (Irrational rotation algebra). Let $A = C(\mathbb{T})$, the continuous functions on the torus, and let $G = \mathbb{Z}$ act by rotation through angle $2\pi\theta$ for some fixed $\theta \in \mathbb{R} \setminus \mathbb{Q}$, i.e.

$$\alpha_n(f)(z) = f\big(e^{-2\pi i n\theta} z\big), \qquad z \in \mathbb{T},\ n \in \mathbb{Z}.$$

The resulting crossed product

$$A_\theta \;:=\; C(\mathbb{T}) \rtimes_\alpha \mathbb{Z}$$

is the *irrational rotation algebra* (a non-commutative torus). It is generated by a unitary u (implementing the $\mathbb{Z}$-action) and a unitary v (generating $C(\mathbb{T})$) subject to the single relation

$$uv = e^{2\pi i\theta}\, vu,$$

and is a standard first example of a simple, non-commutative C^*-algebra arising from a crossed product.

2.3 Matrix Lie Groups and Algebras

In this section we recall some notions from the theory of matrix Lie groups and algebras that will be needed later in the thesis. A *Lie group* is a group that is simultaneously a smooth manifold, in such a way that both the group multiplication and the inversion map are smooth. A *Lie algebra* $\mathfrak{g}$ is a vector space over a field (for us, always $\mathbb{C}$) equipped with a bilinear form, the *Lie bracket*,

$$[\cdot,\cdot]\colon \mathfrak{g}\times\mathfrak{g}\to\mathfrak{g},$$

which is antisymmetric and satisfies the Jacobi identity:

- $[x, y] = -[y, x]$,
- $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

These two notions are intimately related: every Lie group gives rise to a Lie algebra, obtained as the tangent space at the identity element, together with a bracket induced by the group structure.

Our primary interest lies in *matrix Lie groups*, i.e. subgroups of the general linear group

$$GL(n, \mathbb{C}) = \{A \in M_n(\mathbb{C}) : \det(A) \neq 0\},$$

taken with matrix multiplication as the group operation. Important examples include the unitary group

$$U(n) = \{U \in M_n(\mathbb{C}) : U^*U = UU^* = I\},$$

its subgroup, the special unitary group

$$SU(n) = \{U \in M_n(\mathbb{C}) : U^*U = UU^* = I,\ \det(U) = 1\},$$

and the projective unitary group

$$PU(n) = U(n)/Z(U(n)),$$

where $Z(U(n))$ denotes the center of $U(n)$. Each of these groups carries the subspace topology inherited from $M_n(\mathbb{C})$, except for $PU(n)$, which instead carries the quotient topology induced by the projection $U(n) \to U(n)/Z(U(n))$.

Proposition 2.3.1. *$Z(U(n)) \cong \mathbb{T}$.*

Proof. Recall that the center of $U(n)$ is

$$Z(U(n)) = \{A \in U(n) : AU = UA \text{ for all } U \in U(n)\}.$$

Let $A \in Z(U(n))$. Since A commutes with every unitary matrix, it in particular commutes with every permutation matrix. Let P_{ij} denote the permutation matrix exchanging the i-th and j-th basis vectors. From

$$AP_{ij} = P_{ij}A$$

we see that conjugating A by P_{ij} leaves it unchanged. But conjugation by P_{ij} swaps the i-th and j-th rows and columns of A; since this holds for every pair i, j, all off-diagonal entries of A must vanish and all diagonal entries must coincide. Hence $A = \lambda I_n$ for some $\lambda \in \mathbb{C}$, where by I_n we denote the $n \times n$-identity matrix.

As $A \in U(n)$, we have $A^*A = I$, and substituting $A = \lambda I_n$ gives

$$(\overline{\lambda} I_n)(\lambda I_n) = |\lambda|^2 I_n = I_n,$$

so $|\lambda| = 1$.

Conversely, if $A = \lambda I_n$ with $|\lambda| = 1$, then clearly $A \in U(n)$, and for every $U \in U(n)$,

$$AU = \lambda U = UA,$$

so $A \in Z(U(n))$. This proves the claim. □

We will also make use of

$$SO(n) = \{R \in M_n(\mathbb{R}) : R^T R = RR^T = I,\ \det(R) = 1\},$$

the Lie group of orientation-preserving rotations fixing the origin.

For matrix Lie algebras, the Lie bracket is given by the matrix commutator

$$[X, Y] = XY - YX$$

for X, Y in the Lie algebra; one readily checks that this bracket is antisymmetric and satisfies the Jacobi identity.

The Lie algebra associated to $GL(n, \mathbb{C})$ is denoted $\mathfrak{gl}(n, \mathbb{C})$ and consists of *all* complex $n \times n$ matrices,

$$\mathfrak{gl}(n, \mathbb{C}) = M_n(\mathbb{C}).$$

The Lie algebra associated to $SU(n)$, denoted $\mathfrak{su}(n)$, consists of the traceless skew-Hermitian matrices,

$$\mathfrak{su}(n) = \{X \in M_n(\mathbb{C}) : X^* = -X,\ \mathrm{Tr}(X) = 0\}.$$

We may regard $\mathfrak{su}(2)$ as a 3-dimensional real vector space (why?).

Example 2.3.1 (The circle group $SO(2)$ and its Lie algebra). The group $SO(2)$ consists of rotation matrices

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \theta \in [0, 2\pi),$$

and is isomorphic, as a Lie group, to the circle $\mathbb{T}$ via $\theta \mapsto R_\theta$. Differentiating the defining relation $R^T R = I$ at $\theta = 0$ shows that the corresponding Lie algebra $\mathfrak{so}(2)$ consists of antisymmetric real matrices,

$$\mathfrak{so}(2) = \left\{ \begin{pmatrix} 0 & -t \\ t & 0 \end{pmatrix} : t \in \mathbb{R} \right\},$$

a one-dimensional real vector space. Since any two elements of $\mathfrak{so}(2)$ are scalar multiples of one another, the Lie bracket on $\mathfrak{so}(2)$ vanishes identically: it is an *abelian* Lie algebra.

Example 2.3.2 (A basis for $\mathfrak{su}(2)$). A convenient real basis for $\mathfrak{su}(2)$ is given by $\{i\sigma_1, i\sigma_2, i\sigma_3\}$, where $\sigma_1, \sigma_2, \sigma_3$ are the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Each $i\sigma_k$ is traceless and skew-Hermitian, so $i\sigma_k \in \mathfrak{su}(2)$, and the three matrices are linearly independent over $\mathbb{R}$; together with $\dim_{\mathbb{R}} SU(2) = 3$ this confirms that $\mathfrak{su}(2)$ is a 3-dimensional real vector space, as anticipated above. The commutation relations

$$[i\sigma_1, i\sigma_2] = -2i\sigma_3, \qquad [i\sigma_2, i\sigma_3] = -2i\sigma_1, \qquad [i\sigma_3, i\sigma_1] = -2i\sigma_2$$

exhibit $\mathfrak{su}(2)$ as (a rescaling of) the cross-product Lie algebra on $\mathbb{R}^3$, and identify $\mathfrak{su}(2)$ with $\mathfrak{so}(3)$ up to isomorphism.

Example 2.3.3 (Non-example: $GL(n, \mathbb{R})$ inside $GL(n, \mathbb{C})$). The real general linear group

$$GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det(A) \neq 0\}$$

is a matrix Lie group in its own right, and sits inside $GL(n, \mathbb{C})$ as a subgroup, but *not* as a complex Lie subgroup: its associated Lie algebra

$$\mathfrak{gl}(n, \mathbb{R}) = M_n(\mathbb{R})$$

is only a *real* vector space, and is not closed under multiplication by i. This illustrates that subgroups of $GL(n, \mathbb{C})$ need not inherit a complex Lie algebra structure, even though $\mathfrak{gl}(n, \mathbb{C})$ itself is a complex vector space.

Chapter 3

Quantum sets and quantum graphs

In this chapter we will introduce the notions of quantum sets and quantum graphs. They were first considered by...

3.1 Quantum sets

Let B be a C^*-algebra equipped with a faithful positive functional ψ. By the Gelfand–Naimark theorem, B can be realized concretely as a closed $*$-subalgebra of the bounded linear operators $B(\mathcal{H})$ on some Hilbert space $\mathcal{H}$. When B is finite-dimensional, this abstract picture simplifies considerably: rather than passing to $\mathcal{H}$, we can identify B with its own Hilbert space completion $L^2(B, \psi)$, by equipping B itself with the inner product induced by ψ,

$$\langle x, y \rangle = \psi(x^* y), \qquad x, y \in B.$$

Since ψ is faithful, this form is positive definite, and since B is already finite-dimensional, it is automatically complete; hence $L^2(B, \psi)$ may simply be taken to be B itself, viewed as a Hilbert space via this inner product.

This Hilbert space structure is the key extra piece of data that allows one to speak of adjoints of maps built from B, and in particular of the adjoint of multiplication. This leads to the central definition of this thesis, taken directly from Schäfer and Tobolski [9].

Definition 3.1.1 (Quantum set, Schäfer and Tobolski [9], Definition 2.1)**.** () A *quantum set* (B, ψ) is a pair consisting of a finite-dimensional C^*-algebra B and a faithful positive functional $\psi \colon B \to \mathbb{C}$ such that

$$mm^* = \mathrm{Id}_B$$

holds, where $m\colon B \otimes B \to B$ is the multiplication map and m^* is its adjoint with respect to the inner products induced by ψ, i.e. the unique linear map determined by

$$\langle m(a \otimes b), c\rangle = \langle a \otimes b, m^*(c)\rangle_{\psi\otimes\psi}$$

for all $a, b, c \in B$.

Remark 3.1.1. We refer to any functional ψ satisfying the condition in the definition above as a *quantum set functional* on B. The condition $mm^* = \mathrm{Id}_B$ should be thought of as a normalization condition on ψ: not every faithful positive functional on a finite-dimensional C^*-algebra makes (B, ψ) into a quantum set, only those compatible with the multiplication in this precise sense.

Remark 3.1.2. Recall that, more generally, a *comultiplication* on a C^*-algebra A is a $*$-homomorphism

$$\Delta\colon A \to A \otimes A$$

making the following diagram commute (coassociativity):

$$\begin{tikzcd} A \arrow[r, "\Delta"] \arrow[d, "\Delta"'] & A \otimes A \arrow[d, "\Delta\otimes\mathrm{id}"] \\ A \otimes A \arrow[r, "\mathrm{id}\otimes\Delta"'] & A \otimes A \otimes A \end{tikzcd}$$

In general such a Δ is far from unique, and need not interact with any given inner product on A at all. In our setting, however, the situation is much more rigid: once (B, ψ) is a quantum set, the adjoint m^* of the multiplication map is automatically a comultiplication on B, and it is the *unique* one arising this way, since it is pinned down by m together with the Riesz representation theorem for the Hilbert space $L^2(B, \psi)$.

Remark 3.1.3. Because in our setup the comultiplication is obtained from the ordinary multiplication m purely by adjunction with respect to ψ, we will not introduce separate notation for it, and will simply write m^* throughout.

Example 3.1.1 (Classical finite sets as quantum sets). Every finite set V gives rise to a quantum set $(\mathbb{C}^V, \psi_V)$, where $\mathbb{C}^V$ denotes the C^*-algebra of complex-valued functions on V with pointwise multiplication and involution given by pointwise conjugation, $f^*(v) = \overline{f(v)}$, and where

$$\psi_V\colon \mathbb{C}^V \to \mathbb{C}, \qquad \psi_V(f) = \sum_{v\in V} f(v).$$

To see that ψ_V is positive, note that for any $f \in \mathbb{C}^V$ the product f^*f evaluates pointwise as

$$(f^*f)(v) = \overline{f(v)}\, f(v) = |f(v)|^2,$$

so that

$$\psi_V(f^*f) = \sum_{v\in V} |f(v)|^2 \geq 0.$$

For faithfulness, suppose $\psi_V(f^*f) = 0$ for some $f \in \mathbb{C}^V$, i.e.

$$\sum_{v \in V} |f(v)|^2 = 0.$$

Since each summand $|f(v)|^2$ is non-negative, the sum can vanish only if every term does, so $f(v) = 0$ for all $v \in V$, and hence $f = 0$. Thus ψ_V is faithful.

It remains to check the quantum set condition $mm^* = \mathrm{Id}$. Let $\{e_v\}_{v \in V}$ denote the orthonormal basis of $\mathbb{C}^V$ given by the characteristic functions of points, so that

$$m(f \otimes g) = fg, \qquad m^*(e_v) = e_v \otimes e_v.$$

Evaluating the composite mm^* on a basis element gives

$$(mm^*)(e_v) = m\big(m^*(e_v)\big) = m(e_v \otimes e_v) = e_v e_v = e_v,$$

since e_v is idempotent. As mm^* and Id agree on the basis $\{e_v\}_{v \in V}$, they agree on all of $\mathbb{C}^V$ by linearity, so $mm^* = \mathrm{Id}$, and $(\mathbb{C}^V, \psi_V)$ is indeed a quantum set.

Remark 3.1.4. The identification of finite sets and commutative C^*-algebras is inspired by Gelfand's duality theorem, which establishes that every commutative C^*-algebra is isomorphic to the algebra of continuous functions on a compact Hausdorff space. In the finite-dimensional case, this space reduces to a discrete finite set, and the algebra's minimal projections correspond directly to the individual elements of the set. Furthermore, the functional ψ_X serves as a quantum analogue of the counting measure (summing over all elements), while the condition $mm^* = \mathrm{Id}$ ensures the correct normalization and structural compatibility of the multiplication map with this measure, generalizing the discrete Cartesian framework of classical sets to the quantum setting.

Example 3.1.2 (The special case $\mathbb{C}^n$). Taking $V = \{1, \ldots, n\}$ in the previous example recovers the quantum set $(\mathbb{C}^n, \psi_n)$, where ψ_n is determined on the standard basis by $\psi_n(e_i) = 1$ and extends linearly to

$$\psi_n(v) = \sum_{i=1}^{n} v_i, \qquad v \in \mathbb{C}^n.$$

Positivity and faithfulness follow exactly as above: writing $v = \sum_i v_i e_i$ and using orthogonality of the basis,

$$v^*v = \Big(\sum_{i=1}^{n} \overline{v_i}\, e_i\Big)\Big(\sum_{j=1}^{n} v_j e_j\Big) = \sum_{i=1}^{n} \overline{v_i} v_i\, e_i = \sum_{i=1}^{n} |v_i|^2 e_i,$$

so that

$$\psi_n(v^*v) = \sum_{i=1}^{n} |v_i|^2 \psi_n(e_i) = \sum_{i=1}^{n} |v_i|^2 \geq 0,$$

which vanishes precisely when every $v_i = 0$, i.e. when $v = 0$. The quantum set condition again reduces to the computation

$$(mm^*)(e_i) = m(e_i \otimes e_i) = e_i, \qquad i = 1, \ldots, n,$$

which forces $mm^* = \mathrm{Id}$ by linearity.

Example 3.1.3 (Matrix algebras). For any $n \in \mathbb{N}_+$, the pair $(M_n, n\,\mathrm{Tr})$ is a quantum set, where Tr denotes the usual trace on M_n. We will later see that this quantum set arises naturally as a crossed product: writing $\hat{\mathbb{Z}}_n$ for the dual of $\mathbb{Z}_n$ acting on $(\mathbb{C}^n, \psi_n)$ via the standard action, one has

$$(M_n, n\,\mathrm{Tr}) \;\cong\; (\mathbb{C}^n \rtimes \hat{\mathbb{Z}}_n,\; \psi_\rtimes),$$

a fact we prove in 3.1.2. $n\,\mathrm{Tr}$ is chosen instead of simply Tr so that the condition $mm^* = \mathrm{Id}$ is fulfilled.

Lemma 3.1.1 (Comultiplication for matrix algebras)**.** *In the quantum set $(M_n, n\,\mathrm{Tr})$, the comultiplication m^* is given on matrix units by*

$$m^*(e_{ij}) = \sum_{k=1}^{n} e_{ik} \otimes e_{kj},$$

where e_{ij} denotes the matrix with a single 1 in the i-th row and j-th column, extended linearly to all of M_n.

Proof. The proof is a direct computation using two standard facts about matrix units:

1. $e_{ab}e_{cd} = \delta_{bc}\, e_{ad}$;
2. $\langle e_{ij}, e_{kl}\rangle = \mathrm{Tr}(e_{ji}e_{kl}) = \delta_{ik}\,\delta_{jl}$,

where δ_{ij} is the Kronecker delta. In particular, (2) shows that $\{e_{ij}\}_{i,j=1,\dots,n}$ forms an orthonormal basis of M_n with respect to the inner product induced by $n\,\mathrm{Tr}$.

Fix arbitrary basis vectors e_{ab}, e_{cd}. By (1),

$$m(e_{ab} \otimes e_{cd}) = e_{ab}e_{cd} = \delta_{bc}\, e_{ad},$$

and hence, for any basis vector e_{ij},

$$\big\langle m(e_{ab} \otimes e_{cd}),\; e_{ij}\big\rangle = \delta_{bc}\big\langle e_{ad}, e_{ij}\big\rangle = \delta_{bc}\,\delta_{ai}\,\delta_{dj}.$$

On the other hand, pairing $e_{ab} \otimes e_{cd}$ against the proposed adjoint gives

$$\Big\langle e_{ab} \otimes e_{cd},\; \sum_{k=1}^{n} e_{ik} \otimes e_{kj}\Big\rangle = \sum_{k=1}^{n}\langle e_{ab}, e_{ik}\rangle\langle e_{cd}, e_{kj}\rangle = \sum_{k=1}^{n} \delta_{ai}\delta_{bk}\,\delta_{ck}\delta_{dj} = \delta_{ai}\,\delta_{bc}\,\delta_{dj},$$

using that the sum over k collapses to the single term $k = b = c$. The two expressions coincide, so

$$\big\langle m(e_{ab} \otimes e_{cd}), e_{ij}\big\rangle = \big\langle e_{ab} \otimes e_{cd}, m^*(e_{ij})\big\rangle$$

for all basis vectors, and since $\{e_{ij}\}$ is an orthonormal basis, this identifies m^* as claimed on the whole of M_n by linearity. $\square$

3.1.1 Crossed product quantum sets

Schäfer and Tobolski [9] introduced the notion of a *crossed product quantum set*. Suppose we are given a finite-dimensional C^*-algebra $\tilde{B}$ together with a dual group action $\hat{\alpha}\colon \hat{G} \to \mathrm{Aut}(\tilde{B})$, where G is a finite abelian group. One may then form the crossed product

$$B := \tilde{B} \rtimes_{\hat{\alpha}} \hat{G}.$$

If $\tilde{B}$ is equipped with a faithful functional $\tilde{\psi}$ making $(\tilde{B}, \tilde{\psi})$ a quantum set, one can construct a corresponding faithful functional on B making B into a quantum set as well, as follows.

Definition 3.1.2 (Crossed product quantum set, Schäfer and Tobolski [9], Definition 3.1)**.** Let $(\tilde{B}, \tilde{\psi})$ be a quantum set, and let $\hat{\alpha}\colon \hat{G} \to \mathrm{Aut}(\tilde{B}, \tilde{\psi})$ be an action of the dual of a finite abelian group G. The *crossed product quantum set* is defined as

$$(\tilde{B}, \tilde{\psi}) \rtimes_{\hat{\alpha}} \hat{G} := \big(\tilde{B} \rtimes_{\hat{\alpha}} \hat{G},\ \psi_{\rtimes}\big),$$

where $\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}$ is the crossed product C^*-algebra, and $\psi_{\rtimes}$ is the functional given by

$$\psi_{\rtimes}\Big(\sum_{\chi \in \hat{G}} b_\chi u_\chi\Big) = n\, \tilde{\psi}(b_1),$$

with $n = |\hat{G}|$.

The proof that this definition is well posed, i.e. that the resulting object is indeed a quantum set, can be found in Schäfer and Tobolski [9, Proposition 3.4].

Example 3.1.4 (Schäfer and Tobolski [9], Example 3.5)*.* If $\hat{\alpha} = \mathrm{triv}$ is the trivial action, then

$$\tilde{B} \rtimes_{\hat{\alpha}} \hat{G} \;\cong\; \tilde{B} \otimes C(G),$$

via the isomorphism $b_\chi u_\chi \mapsto b_\chi \otimes \tilde{u}_\chi$, where $\tilde{u}_\chi \in C(G)$ denotes the function $g \mapsto \chi(g)$ on G.

Lemma 3.1.2. *Let $\hat{\alpha}\colon \hat{\mathbb{Z}}_n \to \mathrm{Aut}(\mathbb{C}^n)$ be the action given by cyclically shifting coordinates. Then*

$$(M_n, n\,\mathrm{Tr}) \;\cong\; (\mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n,\ \psi_{\rtimes}).$$

Proof. Identify $\hat{\mathbb{Z}}_n$ with $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ (with arithmetic taken modulo n), so that a general element of $\mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n$ is written $b = \sum_{k \in \mathbb{Z}_n} b_k u_k$ with $b_k \in \mathbb{C}^n$, and the action is $\hat{\alpha}_k(e_i) = e_{i+k}$ on the standard basis $\{e_i\}_{i \in \mathbb{Z}_n}$ of $\mathbb{C}^n$, as in the previous examples.

We construct an explicit $*$-isomorphism $\pi\colon \mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n \to M_n$ as follows. On $\mathbb{C}^n$, let M_{b_k} denote the diagonal matrix with entries $b_k(0), \dots, b_k(n-1)$, i.e. $M_{b_k} = \sum_i b_k(i)\, e_{ii}$ in the matrix unit notation of the earlier lemma. Let S be the cyclic shift matrix defined by $Se_i = e_{i+1}$ for the standard basis vectors e_i of $\mathbb{C}^n$ (indices mod n), and set

$$\pi(b) = \sum_{k\in\mathbb{Z}_n} M_{b_k}\, S^k, \qquad b = \sum_{k\in\mathbb{Z}_n} b_k u_k.$$

π **is a $*$-homomorphism.** A direct computation shows that $S^k M_f S^{-k} = M_{\hat\alpha_k(f)}$ for $f \in \mathbb{C}^n$, which is exactly the covariance relation $u_k b u_k^* = \hat\alpha_k(b)$ required of the crossed product. Using this, for $a = \sum_k a_k u_k$ and $b = \sum_l b_l u_l$,

$$\pi(a)\pi(b) = \sum_{k,l} M_{a_k} S^k M_{b_l} S^l = \sum_{k,l} M_{a_k}\, M_{\hat\alpha_k(b_l)}\, S^{k+l} = \sum_m M_{\sum_{k+l=m} a_k \hat\alpha_k(b_l)}\, S^m = \pi(ab),$$

matching the multiplication rule on $C_c(\hat{\mathbb{Z}}_n, \mathbb{C}^n, \hat\alpha)$ from the previous subsection, and similarly

$$\pi(a)^* = \sum_k S^{-k} M_{a_k^*} = \sum_k M_{\hat\alpha_{-k}(a_k^*)}\, S^{-k} = \pi(a^*),$$

matching the involution formula. So π is a $*$-homomorphism.

π **is bijective.** Write $e_i = \delta_i \in \mathbb{C}^n$ for the standard basis vectors. Since S^k sends $e_j \mapsto e_{j+k}$, the operator $M_{e_i} S^k$ sends $e_{i-k} \mapsto e_i$ and annihilates every other basis vector, i.e.

$$\pi(e_i u_k) = M_{e_i} S^k = e_{i,\,i-k},$$

a matrix unit in the sense of the earlier lemma. As i ranges over $\mathbb{Z}_n$ and k ranges over $\mathbb{Z}_n$, the pair $(i, i-k)$ ranges exactly once over all of $\mathbb{Z}_n \times \mathbb{Z}_n$ (for fixed i, varying k makes $i-k$ range over all residues). Hence $\{\pi(e_i u_k)\}_{i,k\in\mathbb{Z}_n}$ is precisely the full set of matrix units $\{e_{ab}\}_{a,b\in\mathbb{Z}_n}$, which forms a basis of M_n. Since $\{e_i u_k\}_{i,k}$ is a basis of $\mathbb{C}^n \rtimes_{\hat\alpha} \hat{\mathbb{Z}}_n$ (of the same dimension n^2) and π carries it bijectively to a basis of M_n, the linear map π is a bijection.

π **intertwines the functionals.** Using $\pi(e_i u_k) = e_{i,i-k}$ and the fact that a matrix unit e_{ab} has trace δ_{ab}, we get

$$\mathrm{Tr}\left(\pi(e_i u_k)\right) = \delta_{i,\,i-k} = \delta_{k,0}.$$

Hence for a general element $b = \sum_k b_k u_k = \sum_{k,i} b_k(i)\, e_i u_k$,

$$\mathrm{Tr}(\pi(b)) = \sum_{k,i} b_k(i)\ \mathrm{Tr}\left(\pi(e_i u_k)\right) = \sum_{k,i} b_k(i)\, \delta_{k,0} = \sum_i b_0(i) = \tilde\psi(b_0),$$

where $\tilde\psi$ is the quantum set functional $\tilde\psi(x) = \sum_i x_i$ on $(\mathbb{C}^n, \tilde\psi)$ from the earlier example, and b_0 is precisely the coefficient at the trivial character (i.e. $k = 0$) appearing in the definition of $\psi_\rtimes$. Multiplying by $n = |\hat{\mathbb{Z}}_n|$,

$$n\,\mathrm{Tr}(\pi(b)) = n\,\tilde\psi(b_0) = \psi_\rtimes(b).$$

Thus π is a $*$-isomorphism $\mathbb{C}^n \rtimes_{\hat\alpha} \hat{\mathbb{Z}}_n \to M_n$ satisfying $n\,\mathrm{Tr}\circ\pi = \psi_\rtimes$, which is exactly an isomorphism of quantum sets

$$(\mathbb{C}^n \rtimes_{\hat\alpha} \hat{\mathbb{Z}}_n,\ \psi_\rtimes)\ \cong\ (M_n,\ n\,\mathrm{Tr}). \qquad \square$$

3.2 Quantum graphs

A graph is classically defined as a pair (V, E) with V serving as the *set of vertices* and $E \subseteq V^2$ serving as the *set of edges.* Such a graph could be interpreted as a matrix $A = [a_{ij}] \in \{0,1\}^{n\times n}$ with $(v_i, v_j) \in E \Leftrightarrow a_{ij} = 1$; such a representation is often called the *adjacency matrix.* The key property of such matrices is that they are invariant with respect to entrywise multiplication. In turn we can use this property to generalise graphs on sets to graphs on quantum sets.

The following definitions are due to Schäfer and Tobolski [9].

Definition 3.2.1 (Quantum adjacency matrix, Schäfer and Tobolski [9], Definition 2.5)**.** Let (B, ψ) be a quantum set. A *quantum adjacency matrix* on this quantum set is a linear map $A\colon B \to B$ which is Schur-idempotent and *-preserving, by which we mean that it satisfies

$$m(A \otimes A)m^* = A, \qquad A(b^*) = A(b)^* \quad \text{for all } b \in B.$$

Definition 3.2.2 (Quantum graph, Schäfer and Tobolski [9], Definition 2.5)**.** A *quantum graph* on (B, ψ) is given by a quantum adjacency matrix on this quantum set.

For all intents and purposes we will identify both notions and simply call an appropriate $A\colon B \to B$ both a quantum adjacency matrix and a quantum graph.

Remark 3.2.1. There are other equivalent definitions of a quantum graph, namely those based on

- an *edge projection*, i.e. a projection $P \in B \otimes B^{\mathrm{op}}$, see Musto, Reutter, and Verdon [8],
- operator systems $S \subseteq B(\mathcal{H})$, i.e. linear subspaces of of operators on a Hilbert space $\mathcal{H}$ such that $S = S^*$ and $\mathrm{id}_{\mathcal{H}} \in S$, see Weaver [11],
- linear subspaces $V \subseteq M_3$, see Gromada [3].

We will make use of the latter definition later in the thesis.

Definition 3.2.3 (Schäfer and Tobolski [9], Definition 4.6)**.** Let A be a quantum graph on a quantum set (B, ψ). We call the quantum graph

- *loop-free* if $m(A \otimes \mathrm{Id})m^* = 0$,
- *undirected* if $A = A^*$, where the adjoint is taken with respect to the inner product induced by ψ.

Remark 3.2.2. The motivation behind the first point in the definition above will become clear thanks to 3.3.1, which says that $m(A \otimes B)m^*$ can be thought of as taking the intersection of two graphs edgewise. Subsequently $m(A \otimes \mathrm{Id})m^*$ would keep only loops, i.e. edges going from a vertex to itself. The second point is easy to motivate in a similar manner: a graph is undirected if and only if its adjacency matrix is symmetric.

3.3 Voltage and Derived Quantum Graphs

In this section we introduce the notions of voltage and derived quantum graphs. Voltage graphs and their derived graphs were first introduced by Gross [5]; these notions have since been extended to the setting of quantum graphs by Schäfer and Tobolski [9].

3.3.1 Classical voltage graphs

Before turning to the quantum setting, it is instructive to first recall the classical construction. Informally, a voltage graph is simply a graph whose edges have been labeled, or *weighted*, by elements of a finite group; the derived graph then unfolds this labeled graph into a (typically much larger) covering graph, built by tracking how these group elements accumulate as one walks along paths in the original graph. This construction originates in topological graph theory, where it was used to build graph embeddings on surfaces of higher genus, but it also turns out to provide a rich and flexible source of examples of graphs with interesting symmetry, which is the perspective we adopt here.

Definition 3.3.1 (Classical voltage graph)**.** A *voltage graph* is a pair (Γ, λ) consisting of a (not necessarily simple) finite directed graph Γ, with vertex set V and edge set E, together with a labelling $\lambda\colon E \to G$ of the edges of Γ by elements of a finite group G.

From every voltage graph one can construct a new graph, called its *derived graph*, as follows.

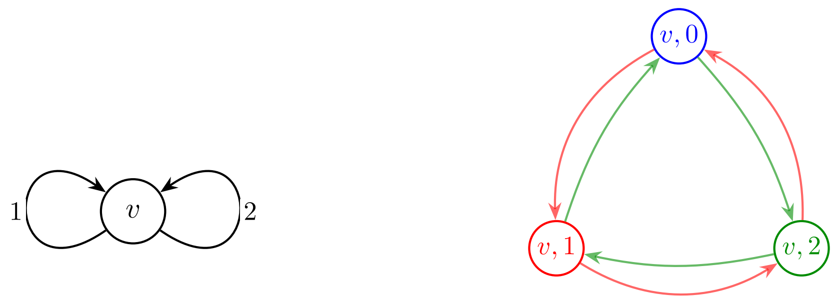
Definition 3.3.2 (Classical derived graph)**.** The *derived graph* Γ^λ of (Γ, λ) has vertex set $V \times G$ and edge set $E \times G$. Whenever $e \in E$ is an edge from u to v in Γ, and $g \in G$ is a group element, the corresponding edge (e, g) in Γ^λ runs from (u, g) to $(v,\, g\lambda(e))$.

Intuitively, the derived graph consists of $|G|$ labeled copies (*fibers*) of the vertex set of Γ, one for each group element, and every edge e of Γ is lifted to $|G|$ edges in Γ^λ, one leaving each fiber, whose endpoint is determined by translating within the target fiber according to the voltage $\lambda(e)$. This is precisely the picture underlying the worked examples given below.

The following is a major result in the theory of voltage and derived graphs, showing that this construction is not merely a convenient way of producing examples, but in fact accounts for *every* graph admitting a free action of a finite group. It can be found in Gross and Tucker [4, Theorem 2.2.2].

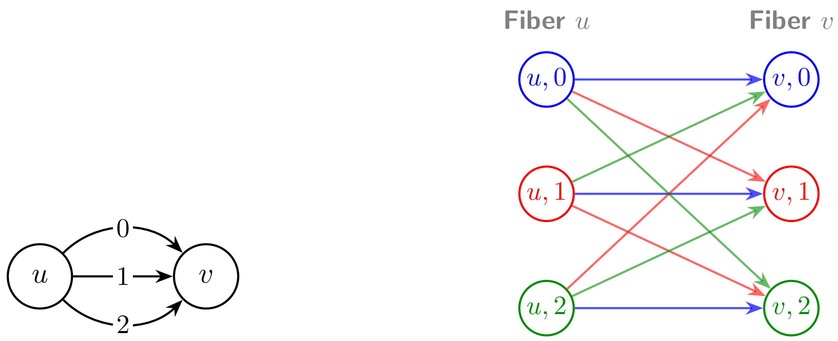
Theorem 3.3.1 (Gross and Tucker theorem)**.** *Let $\tilde{\Gamma}$ be a graph and let G be a finite group acting freely on $\tilde{\Gamma}$ by automorphisms. Then there exists a voltage graph (Γ, λ), with edges labeled by elements of G, such that $\tilde{\Gamma}$ is isomorphic to the derived graph of (Γ, λ). The graph Γ is the quotient graph $\tilde{\Gamma}/G$.*

In other words, voltage graphs classify free group actions on graphs up to isomorphism: recovering (Γ, λ) from $\tilde{\Gamma}$ amounts to nothing more than quotienting by the group action, while reconstructing $\tilde{\Gamma}$ from (Γ, λ) is exactly the derived graph construction above. Below we give some examples of voltage graphs together with their derived graphs.



Base voltage graph Γ over $\mathbb{Z}_3$ Derived graph Γ^λ (two 3-cycles)

Figure 3.1: Voltage graph and derived cover over $\mathbb{Z}_3$, for a base graph with a single vertex and two loops.



Base theta graph Γ with edge labels Derived graph Γ^λ ($K_{3,3}$ colored by edge class)

Figure 3.2: Voltage graph and derived cover for a theta graph over $\mathbb{Z}_3$.

3.3.2 Generalisation to Quantum Voltage and Derived Quantum Graphs

We now turn to generalising the Gross–Tucker construction to the setting of quantum graphs. This generalisation was first carried out by Bjorn and Tobolski in Schäfer and Tobolski [9], building on the notion of a crossed product quantum set introduced earlier in this chapter. Just as a classical voltage graph is a labelling of the edges of a graph by group elements, a voltage *quantum* graph will be a family of quantum adjacency matrices indexed by the group G rather than a single labelling function; the derived quantum graph then reassembles this family into a single quantum adjacency matrix on the crossed product quantum set $(B, \psi_{\rtimes})$, playing the role of the derived graph Γ^{λ} from the classical picture. The following definition makes this precise.

Definition 3.3.3 (Schäfer and Tobolski [9], Definition 4.1)**.** Suppose $(\tilde{B}, \tilde{\psi})$, the action $\hat{\alpha}\colon \hat{G} \to \mathrm{Aut}(\tilde{B}, \tilde{\psi})$, the crossed product $(B, \psi_{\rtimes})$, and the elements $(u_\chi)_{\chi \in \hat{G}} \subset B$ are as introduced above.

1. For each $g \in G$, set
$$X_g := \frac{1}{n} \sum_{\chi \in \hat{G}} \overline{\chi(g)}\, u_\chi u_\chi^{\dagger},$$
i.e. let X_g be the linear operator
$$X_g \colon B \to B, \qquad b \mapsto \frac{1}{n} \sum_{\chi \in \hat{G}} \overline{\chi(g)}\, \langle u_\chi, b \rangle_{\psi_{\rtimes}}\, u_\chi.$$

2. Given the group action $\hat{\alpha}\colon \hat{G} \to \mathrm{Aut}(\tilde{B}, \tilde{\psi})$, a *voltage quantum graph* on $(\tilde{B}, \tilde{\psi})$ is a G-indexed family $\tilde{A} = (\tilde{A}_g)_{g \in G}$ of quantum adjacency matrices such that
$$\tilde{A}_g \hat{\alpha}_\chi = \hat{\alpha}_\chi \tilde{A}_g$$
holds for every $g \in G$ and every $\chi \in \hat{G}$.

3. Given such a family $\tilde{A}$, the associated *derived quantum graph* $\tilde{A} \rtimes_{\hat{\alpha}} \hat{G} \colon B \to B$ on $(B, \psi_{\rtimes})$ is
$$\tilde{A} \rtimes_{\hat{\alpha}} \hat{G} := \sum_{g \in G} m\big(X_g \otimes E_{\rtimes}^{*} \tilde{A}_g E_{\rtimes}\big)\, m^{*},$$
where $m = m_{\rtimes}$ is the multiplication map on B.

The role played by X_g here is directly analogous to the indicator function of the fiber $V \times \{g\}$ in the classical derived graph: it isolates, within B, the *copy of* g sitting inside the crossed product. The fact that X_g is itself an adjacency matrix on the quantum set $(B, \psi_{\rtimes})$ is checked in Schäfer and Tobolski [9, Proposition 4.5], and the derived quantum graph is then built by combining these fiber projections with the individual voltage components $\tilde{A}_g$, summed over all $g \in G$ – mirroring how, in the classical construction, each edge of Γ^λ is assigned both to a specific fiber and to a specific lift of an edge of Γ.

As in the classical case, it will be useful to know when a voltage quantum graph gives rise to a derived quantum graph with no loops, or one that is undirected.

Definition 3.3.4. A voltage quantum graph $\tilde{A} = \{\tilde{A}_g\}_{g\in G}$ is called *loop-free* or *directed* if the respective properties from Definition 3.2.3 hold for the sum $\sum_{g\in G} \tilde{A}_g$.

We can connect the notions of voltage and derived graphs to their quantum equivalents. The following proposition tells us that classical voltage graphs can be thought of as special kinds of voltage quantum graphs. Below $(\mathbb{C}_V, \psi_V)$ is used to denote a quantum set derived from a finite classical set V, like it was done in Example 3.1.1.

Proposition 3.3.1 (Schäfer and Tobolski [9], Proposition 4.10)**.** *Let G be a finite abelian group, and let $(\mathbb{C}^V, \psi_V)$ be a quantum set. Given a pre-simple voltage graph (Γ, λ) on V with labelling $\lambda\colon E \to G$, let $\tilde{A}_g\colon \mathbb{C}^V \to \mathbb{C}^V$ denote the adjacency matrix of the (simple, directed) subgraph of Γ consisting of exactly those edges labeled g. Then the assignment*

$$(\Gamma, \lambda) \longmapsto \tilde{A} = (\tilde{A}_g)_{g\in G}$$

is a one-to-one correspondence between G-labeled pre-simple voltage graphs on n and voltage quantum graphs on $(\mathbb{C}^V, \psi_V)$ with respect to the trivial action on $(\mathbb{C}^V, \psi_V)$.

In the same manner classical derived graphs can be identified with certain derived quantum graphs. Note that the group action in question is trivial, which connects to Theorem 3.3.1, which states that a graph is isomorphic to a derived graph of some voltage graph in and only if it allows a free group action of some finite group.

Proposition 3.3.2 (Schäfer and Tobolski [9], Proposition 4.12)**.** *In the setting of Proposition 3.3.1, let (Γ, λ) be a pre-simple voltage graph with associated voltage quantum graph $\tilde{A} = (\tilde{A}_g)_{g\in G}$. Then the derived graphs agree under the identification*

$$\Gamma^\lambda = \tilde{A} \rtimes_{\mathrm{triv}} \hat{G},$$

where Γ^λ is the Gross–Tucker derived graph, and $\tilde{A} \rtimes_{\mathrm{triv}} \hat{G}$ is the derived quantum graph of Definition 3.3.3; here classical graphs on $V \times G$ are identified with quantum graphs on $(\mathbb{C}^V, \psi_V) \rtimes_{\mathrm{triv}} \hat{G}$.

Thus we can identify classical derived graphs with quantum graphs on cross-product quantum sets with the trivial action. This will come useful later as this will allow us to establish a quantum isomorphism between nonclassical graphs on $(M_3, 3\,\mathrm{Tr})$ and classical graphs that happen to be derived graphs.

The lemma below lets us better understand the correspondence between classical and quantum derived graphs.

Lemma 3.3.1 (Schäfer and Tobolski [9], Lemma 4.11)**.** *Let $(\mathbb{C}^V, \psi_V)$ be the quantum set associated with a finite set V, and let G be a finite abelian group equipped with the trivial action* $\mathrm{triv}\colon \hat{G} \to \mathrm{Aut}(\mathbb{C}^V, \psi_V)$*. Then following hold.*

1. *The assignment on generators*

$$e_v \mapsto \sum_{g \in G} e_{v,g}, \qquad u_\chi \mapsto \sum_{v \in V,\, g \in G} \chi(g)\, e_{v,g}$$

extends to an isomorphism

$$(\mathbb{C}^V, \psi_V) \rtimes_{\mathrm{triv}} \hat{G} \;\cong\; (\mathbb{C}^{V \times G}, \psi_{V \times G}),$$

where the right-hand side is the quantum set associated to the finite set $V \times G$ as in Example 3.1.1, and e_v, $e_{v,g}$ denote the indicator functions in $\mathbb{C}^V$ and $C(V \times G)$, respectively.

2. *Under this identification, the operator $X_g\colon B \to B$, $g \in G$, from Definition 3.3.3 corresponds to the classical graph on $V \times G$ whose edges are exactly*

$$(v, h) \to (w, hg), \qquad v, w \in V,\; h \in G.$$

3. *Likewise, the operator $E_\rtimes^* \tilde{A}_g E_\rtimes \colon B \to B$ from Definition 3.3.3 corresponds to the classical graph on $V \times G$ whose edges are exactly*

$$(v, h) \to (w, k), \qquad h, k \in G,\; v \xrightarrow{\tilde{A}_g} w,$$

where $v \xrightarrow{\tilde{A}_g} w$ means that v and w are joined by an edge in the classical graph corresponding to the adjacency matrix $\tilde{A}_g$.

4. *Finally, suppose A_1, A_2 are quantum adjacency matrices corresponding, in the sense of Example 3.1.1, to classical graphs Γ_1, Γ_2 on $V \times G$. Then*

$$A_3 := m(A_1 \otimes A_2)\, m^*$$

is again a quantum adjacency matrix, and corresponds to the classical graph $\Gamma_3 := \Gamma_1 \cap \Gamma_2$ with vertex set $V \times G$ and edge set given by the intersection of the edge sets of Γ_1 and Γ_2.

Example 3.3.1 (Schäfer and Tobolski [9], Example 4.13)*.* Consider the voltage graph (Γ, λ) consisting of two vertices v, w along with a single edge e going from v to w and labeled with $1_{\mathbb{Z}_3}$, i.e. $\lambda(e) = 1_{\mathbb{Z}_3}$. The derived graph then has 6 vertices, 3 for each vertex of Γ.

Base voltage graph Γ Derived graph Γ^λ

Figure 3.3: Base voltage graph and its derived cover over $\mathbb{Z}_3$.

As discussed above, a classical graph on n vertices corresponds to a quantum graph on the quantum set $(\mathbb{C}^n, \psi_n)$. The voltage graph (Γ, λ) is thus identified with a voltage quantum graph on $(\mathbb{C}, \psi_1)$, given by a family $(\tilde{A}_g)_{g\in\mathbb{Z}_3}$ of quantum adjacency matrices on $(\mathbb{C}, \psi_1)$. Since Γ has only a single edge, labeled by $1_{\mathbb{Z}_3}$, it is straightforward to check that (Γ, λ) corresponds to the family $\tilde{A} = (\tilde{A}_g)_{g\in\mathbb{Z}_3}$ given by

$$\tilde{A}_{0_{\mathbb{Z}_3}} = 0, \qquad \tilde{A}_{1_{\mathbb{Z}_3}} = \mathrm{id}_{\mathbb{C}}, \qquad \tilde{A}_{2_{\mathbb{Z}_3}} = 0.$$

In general one would need to verify that $(\tilde{A}_g)_{g\in\mathbb{Z}_3}$ satisfies the conditions of Definition 3.3.3 in order to qualify as a voltage quantum graph; here, however, we are regarding $(\tilde{A}_g)_{g\in\mathbb{Z}_3}$ as a voltage quantum graph with respect to the trivial action triv of $\hat{\mathbb{Z}}_3$ on $\mathbb{C}$, for which there is nothing left to check (see Proposition 3.3.1).

What, then, is the derived quantum graph $\tilde{A} \rtimes_{\mathrm{triv}} \hat{\mathbb{Z}}_3$ of this voltage quantum graph? Recall first that the derived quantum graph lives on the crossed product quantum set

$$(\mathbb{C}, \psi_1) \rtimes_{\mathrm{triv}} \hat{\mathbb{Z}}_3.$$

By part (a) of Lemma 3.3.1, this quantum set is isomorphic to $(\mathbb{C}^3, \psi_3)$, and quantum graphs on $(\mathbb{C}^3, \psi_3)$ correspond to classical graphs on three vertices.

By Definition 3.3.3, the derived quantum graph $A := \tilde{A} \rtimes_{\mathrm{triv}} \hat{\mathbb{Z}}_3 \colon \mathbb{C}^3 \to \mathbb{C}^3$ on $(\mathbb{C}^3, \psi_3)$ is given by

$$A = \sum_{g\in\mathbb{Z}_3} m\big(X_g \otimes E_{\rtimes}^* \tilde{A}_g E_{\rtimes}\big)\, m^*.$$

By parts (b) and (c) of Lemma 3.3.1, the quantum graphs X_g and $E_{\rtimes}^* \tilde{A}_g E_{\rtimes}$ correspond to the classical graphs on $\{v_0, v_1, v_2\}$ shown in Figure 3.4 and Figure 3.5, respectively.

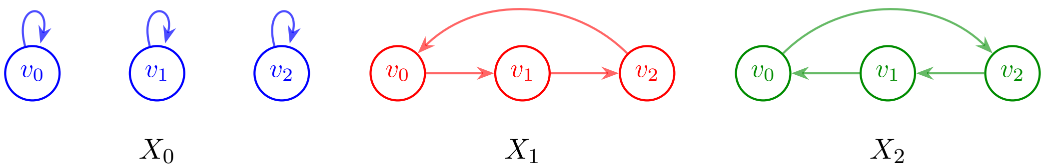


Figure 3.4: The graphs corresponding to the adjacency matrices X_g.

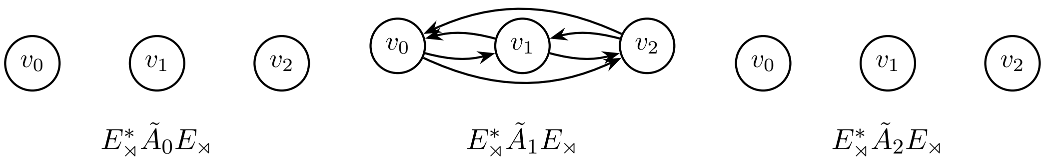


Figure 3.5: The graphs corresponding to the adjacency matrices $E_{\rtimes}^* \tilde{A}_g E_{\rtimes}$.

By part (d) of Lemma 3.3.1, the term

$$m\big(X_g \otimes E_{\rtimes}^* \tilde{A}_g E_{\rtimes}\big)\, m^*$$

returns the adjacency matrix of the classical graph obtained by intersecting the graphs of Figure 3.4 and Figure 3.5. Only the term for $g = 1_{\mathbb{Z}_3}$ contributes to the derived quantum graph A, so that

$$A = m\big(X_1 \otimes E_{\rtimes}^{*}\tilde{A}_1 E_{\rtimes}\big)\, m^{*},$$

and it is straightforward to check that this is exactly the derived graph Γ^{λ} from Figure 3.3.

This confirms, in the simplest possible case, the general correspondence of Proposition 3.3.1 and Proposition 3.3.2: starting from a classical voltage graph, treating it as a voltage quantum graph with respect to the trivial action reproduces exactly the classical derived graph via Definition 3.3.3.

3.4 Isomorphisms and Quantum Isomorphisms of Quantum Sets

Quantum isomorphisms were first conceived in Atserias et al. [1] in the context of game theory. We follow the definitions due to Schäfer and Tobolski [9], which in turn build on the work of Musto, Reutter, and Verdon [8].

Definition 3.4.1 (Quantum isomorphisms, Schäfer and Tobolski [9], Definition 2.8)**.** Let (B_1, ψ_1) and (B_2, ψ_2) be quantum sets, and let $A_1 \colon B_1 \to B_1$ and $A_2 \colon B_2 \to B_2$ be quantum graphs on (B_1, ψ_1) and (B_2, ψ_2), respectively.

- The quantum sets (B_1, ψ_1) and (B_2, ψ_2) are called *isomorphic* if there exists a $*$-isomorphism $\phi \colon B_1 \to B_2$ satisfying $\psi_2 \circ \phi = \psi_1$. The group of all such self-isomorphisms of a quantum set (B, ψ) is denoted $\mathrm{Aut}(B, \psi)$.

- Two quantum graphs are called *isomorphic* if the underlying quantum sets admit an isomorphism $\phi \colon B_1 \to B_2$ which additionally intertwines the graphs, i.e. satisfies $\phi A_1 = A_2 \phi$. The corresponding group of automorphisms of a quantum graph A is written $\mathrm{Aut}(A)$.

- The quantum sets (B_1, ψ_1) and (B_2, ψ_2) are said to be *quantum isomorphic*, denoted $(B_1, \psi_1) \cong_q (B_2, \psi_2)$, if there exist a finite-dimensional Hilbert space $\mathcal{H}$ together with a unital $*$-homomorphism

  $$\rho \colon B_1 \to B_2 \otimes B(\mathcal{H})$$

  for which the induced map

  $$p \colon L^2(B_1) \otimes \mathcal{H} \to L^2(B_2) \otimes \mathcal{H}, \qquad a \otimes \xi \longmapsto \rho(a)(1 \otimes \xi),$$

  is a unitary operator between Hilbert spaces.

- Two quantum graphs A_1, A_2 are *quantum isomorphic*, written $A_1 \cong_q A_2$, if the underlying quantum sets admit a quantum isomorphism $\rho\colon B_1 \to B_2 \otimes B(\mathcal{H})$ that in addition satisfies the intertwining condition

$$\rho A_1 = (A_2 \otimes \mathrm{Id}_{B(\mathcal{H})})\, \rho.$$

Remark 3.4.1. Taking $\mathcal{H} = \mathbb{C}$ in the definition above recovers the ordinary notion of isomorphism as a special case: a unital $*$-homomorphism $\rho\colon B_1 \to B_2 \otimes B(\mathbb{C}) = B_2$ for which $p = \rho$ is unitary is exactly a $*$-isomorphism $\phi\colon B_1 \to B_2$ satisfying $\psi_2 \circ \phi = \psi_1$, and the intertwining condition $\rho A_1 = (A_2 \otimes \mathrm{Id})\rho$ reduces to $\phi A_1 = A_2 \phi$. In this sense, quantum isomorphism genuinely generalizes the classical notion above: it replaces a single isomorphism ϕ between the quantum sets themselves by a family of maps, twisted by an auxiliary Hilbert space $\mathcal{H}$, that need only behave like an isomorphism *on average* over $\mathcal{H}$. This is exactly what allows two quantum graphs to be quantum isomorphic without being isomorphic in the ordinary sense, mirroring how, in the game-theoretic origin of the notion, a quantum strategy can win a non-local game that no classical strategy can win.

Quantum isomorphism is thus a strictly weaker notion than ordinary isomorphism: every isomorphism of quantum sets or quantum graphs is in particular a quantum isomorphism, but not conversely. Later, we will use this weaker notion to establish quantum isomorphisms between certain classical derived graphs and quantum graphs on $(M_3, 3\,\mathrm{Tr})$ that are not, and cannot be, isomorphic in the ordinary sense.

The theorem below lets us establish quantum isomorphisms between derived quantum graphs differing only by the choice of group action.

Theorem 3.4.1 (Schäfer and Tobolski [9], Theorem 5.2)**.** *Let $(\tilde{B}, \tilde{\psi})$, the action $\hat{\alpha}\colon \hat{G} \to \mathrm{Aut}(\tilde{B}, \tilde{\psi})$, and the crossed product $(\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}, \psi_{\rtimes})$ be as introduced above, and let $\tilde{A} = (\tilde{A}_g)_{g \in G}$ be a voltage quantum graph on $(\tilde{B}, \tilde{\psi})$ with respect to the action $\hat{\alpha}$ of $\hat{G}$. Then the corresponding derived quantum graphs satisfy the quantum isomorphism*

$$\tilde{A} \rtimes_{\hat{\alpha}} \hat{G} \;\cong_q\; \tilde{A} \rtimes_{\mathrm{triv}} \hat{G}.$$

Proof. See Schäfer and Tobolski [9] for details. □

Theorem 3.4.1 will be central to our strategy in Section 4.3 by choosing $\hat{\alpha}$ to be the trivial action, we already know from Proposition 3.3.1 and Lemma 3.3.1 that $\tilde{A} \rtimes_{\mathrm{triv}} \hat{G}$ is (isomorphic to) a *classical* derived graph. The theorem then lets us transport this classical realization back along any other action $\hat{\alpha}$ we might prefer to work with, establishing a quantum isomorphism between the (possibly non-classical) derived quantum graph $\tilde{A} \rtimes_{\hat{\alpha}} \hat{G}$ and a genuinely classical graph, without having to construct that isomorphism by hand in each individual case. This is precisely the tool we will use to show that every one of the five equivalence classes of loop-free, undirected quantum graphs on $(M_3, 3\,\mathrm{Tr})$ established in Corollary 4.2.2 is quantum isomorphic to a classical graph on nine vertices.

3.5 Gromada's theorem

In this section we state a theorem due to Gromada [3] establishing a one-to-one correspondence between certain linear subspaces of $\mathfrak{sl}_n$ and quantum graphs. This result will be used to classify all loop-free and undirected quantum graphs on M_3 up to quantum isomorphism, and it also provides an explicit recipe for recovering the adjacency matrix corresponding to a given subspace.

Theorem 3.5.1 (Gromada [3], Theorem 3.7)**.** *Let $m \in \{0, 1, \ldots, n^2\}$. Quantum graphs on M_n with m quantum edges are in one-to-one correspondence with m-dimensional subspaces $V \subseteq \mathfrak{gl}_n$, the adjacency matrix being given by*

$$A_V = \sum_{s=1}^{m} \xi_s \xi_s^*,$$

where $(\xi_1, \ldots, \xi_m)$ is any orthonormal basis of V.

The quantum graph has no loops if and only if $V \subseteq \mathfrak{sl}_n$, and is undirected if and only if $V = V^*$.

If $V, W \subseteq \mathfrak{gl}_n$ correspond to isomorphic quantum graphs, then $V = UWU^*$ for some $U \in SU_n$.

Before applying this correspondence, it will be convenient to have a more concrete handle on subspaces satisfying $V = V^*$, since this is precisely the condition singling out undirected quantum graphs. The next two lemmas establish that such subspaces always admit a basis of self-adjoint elements, and that this self-adjoint data is in fact enough to reconstruct V uniquely.

Lemma 3.5.1 (Gromada [3], Lemma 3.8)**.** *Every vector subspace $V \subseteq \mathfrak{gl}_n$ with $V = V^*$ admits a self-adjoint basis $\{\xi_s\}$, i.e. with $\xi_s = \xi_s^*$ for every s.*

Proof. We construct the desired basis inductively. Suppose we have already obtained a linearly independent set of self-adjoint vectors $\{\xi_s\}$ spanning a subspace $V_1 \subseteq V$. By construction, V_1 is invariant under the involution, i.e. $V_1 = V_1^*$.

If $V_1 \neq V$, let $V_2 = V_1^{\perp}$ be the orthogonal complement of V_1 with respect to the inner product. Since the inner product and the involution are compatible, V_2 is also invariant under $\cdot^*$.

Choose any nonzero $\xi \in V_2$ and decompose it into its self-adjoint real and imaginary parts,

$$\xi_{\mathrm{Re}} = \frac{\xi + \xi^*}{2}, \qquad \xi_{\mathrm{Im}} = \frac{i(\xi^* - \xi)}{2}.$$

Both ξ_{Re} and ξ_{Im} are self-adjoint elements of V_2, and since $\xi \neq 0$, at least one of them is nonzero. We extend our independent set by appending either both (if linearly independent) or the single nonzero one (if not). Iterating this process eventually produces a full self-adjoint basis of V. $\square$

The following lemma shows that any undirected quantum graph on M_n may be thought of as being built out of symmetric quantum edges alone.

Lemma 3.5.2 (Gromada [3], Remark 3.10)**.** *A complex vector space $V = V^* \subseteq \mathfrak{gl}_n$ uniquely determines a real vector space, spanned by a basis $\{\xi_s\}$ of self-adjoint elements $\xi_s = \xi_s^*$.*

Proof. Let $V \subseteq \mathfrak{gl}_n$ be a complex vector space with $V = V^*$.

Since V is invariant under the Hermitian adjoint, every $v \in V$ admits the decomposition

$$v = \frac{v + v^*}{2} + i\,\frac{v - v^*}{2i}.$$

Both summands lie in V, as V is closed under addition, scalar multiplication, and the adjoint operation, and moreover

$$\left(\frac{v + v^*}{2}\right)^* = \frac{v + v^*}{2}, \qquad \left(i\,\frac{v - v^*}{2i}\right)^* = i\,\frac{v - v^*}{2i},$$

so both are self-adjoint.

Now let $\xi \in V$ be self-adjoint, and let $\{\xi_s\}$ be a self-adjoint basis of V. Write $\xi = \sum_s c_s \xi_s$ for some $c_s \in \mathbb{C}$. Taking the Hermitian adjoint gives

$$\xi = \xi^* = \sum_s \overline{c_s}\, \xi_s,$$

since each ξ_s is self-adjoint, and by uniqueness of the expansion with respect to $\{\xi_s\}$ we get $c_s = \overline{c_s}$ for every s, i.e. every coefficient c_s is real.

Hence every self-adjoint element of V is a real linear combination of $\{\xi_s\}$, so the self-adjoint part of V is a real vector space uniquely determined by V. $\square$

By the above lemma, we may regard any self-adjoint $V \subseteq \mathfrak{sl}_n$ as a real subspace of $\mathfrak{sl}_n$.

Remark 3.5.1 (Gromada [3], Section 2.3)*.* For a quantum set (B, ψ), it is tempting to call the minimal central projections of B (i.e. the matrix blocks M_{n_i} in a decomposition $B \cong \bigoplus_i M_{n_i}$) its *vertices.* This works well enough for classical sets, where $B = \mathbb{C}^n$ decomposes into n copies of $M_1 = \mathbb{C}$, recovering the usual n vertices. However, for a genuinely noncommutative quantum set such as $(M_n, n\,\mathrm{Tr})$, this convention is misleading: it would count M_n as a single *vertex*, even though $\dim(M_n) = n^2$ and the quantum graphs living on M_n behave, in every respect relevant to us, like graphs on n^2 points rather than on a single point.

This is exactly why, in Theorem 3.5.1, the number of *quantum edges* m of a quantum graph on M_n is counted as $\dim(V)$ for the corresponding subspace $V \subseteq \mathfrak{gl}_n$, rather than by some more naive combinatorial count: it is the dimension of the underlying vector space, not a literal count of *vertices* or *edges* in any classical sense, that behaves correctly under quantum isomorphism and yields consistent statements such as Corollary 4.2.1 and the classification of Table 4.1. We will consistently use m, i.e. the dimension of the relevant subspace, as our notion of *size* for a quantum graph throughout the next chapter.

3.6 Summary

This chapter laid the algebraic groundwork for treating classical combinatorial structures and their quantum analogues on equal footing. The guiding principle throughout has been that a finite set, once recast as a commutative C^*-algebra with a suitable functional, admits a natural non-commutative generalization – the quantum set – and that many further classical notion built on top of a set generalizes correspondingly. We saw this principle realized concretely: matrix algebras arose as crossed products of classical quantum sets, classical voltage and derived graphs embedded as a special (trivial-action) case of their quantum counterparts, and Gromada's correspondence translated quantum graphs on M_n into linear-algebraic data – subspaces of $\mathfrak{gl}_n$ – amenable to direct computation. Quantum isomorphisms, which are a generalisation of regular isomorphisms, will help us relate classical graphs to purely quantum ones in the next chapter.

Figure 3.6 summarizes this dictionary between the classical and quantum sides, for reference throughout the remainder of the thesis.

classical concept	quantum generalization
finite set	quantum set (B, ψ)
bijection	quantum set isomorphism
directed graph, adjacency matrix A	quantum graph, subspace $V \subseteq \mathfrak{gl}_n$
loop-free graph	$V \subseteq \mathfrak{sl}_n$
undirected graph	self-adjoint subspace
graph isomorphism	$V_1 = UV_2U^*$ for some $U \in U(n)$
group action	(dual) group action $\hat{\alpha} \colon \hat{G} \to \mathrm{Aut}(B, \psi)$
???	crossed product quantum set $\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}$
voltage graph (Γ, λ)	voltage quantum graph $\tilde{A} = (\tilde{A}_g)_{g \in G}$
derived graph Γ^λ	derived quantum graph $\tilde{A} \rtimes_{\hat{\alpha}} \hat{G}$
intersection of graphs	$m(A_1 \otimes A_2)m^*$
isomorphism	quantum isomorphism

Figure 3.6: A dictionary between classical combinatorial notions and their quantum counterparts, as developed throughout this thesis.

Chapter 4

Quantum graphs over 3×3 matrices

The classification of quantum graphs on 2×2 matrices was carried out independently by Matsuda [7] and Gromada [3]. In this chapter we attempt a partial classification of quantum graphs on 3×3 matrices. This case turns out to be considerably harder than the 2×2 case, and a classification of comparable simplicity is likely out of reach altogether, as we show in Lemma 4.5.1. We conclude the chapter by comparing our results with those of Hayes et al. [6].

4.1 Pauli matrices and their generalisations

4.1.1 Pauli matrices

An important basis of the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ is given by the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Pauli matrices are Hermitian, traceless, and linearly independent; since $\dim_{\mathbb{C}} \mathfrak{sl}(2, \mathbb{C}) = 3$, they form a basis of $\mathfrak{sl}(2, \mathbb{C})$. They were already encountered in Example 2.3.2.

Because the Pauli matrices are Hermitian rather than skew-Hermitian, they do not themselves belong to $\mathfrak{su}(2)$. Instead, it is the matrices

$$i\sigma_1, \qquad i\sigma_2, \qquad i\sigma_3$$

that form a basis of $\mathfrak{su}(2)$, being both traceless and skew-Hermitian.

4.1.2 Generalised Pauli matrices

In dimensions greater than two, the role of the Pauli matrices is played by the *generalised Pauli matrices*, also known as the *clock* and *shift* matrices. In dimension three, they are defined by

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

where $\omega = \exp(2\pi i/3)$.

The matrices P and Q are unitary and satisfy

$$P^3 = Q^3 = I,$$

together with the commutation relation

$$PQ = \omega\, QP.$$

The clock and shift matrices are related by the discrete Fourier transform. The Fourier matrix is

$$F = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix},$$

which is unitary, i.e. satisfies $F^*F = FF^* = I$, and exchanges the roles of the clock and shift operators according to

$$FPF^* = Q^{-1}, \qquad FQF^* = P.$$

Consequently, P and Q are unitarily equivalent and represent two complementary bases of the three-dimensional Hilbert space. This duality is the finite-dimensional analogue of the relation between position and momentum operators, and is a fundamental feature of the generalised Pauli matrices.

Lemma 4.1.1. *The matrices P^aQ^b, $a, b \in \{0, 1, 2\}$, form an orthogonal basis of M_3 with respect to the Hilbert–Schmidt inner product $\langle A, B\rangle = \mathrm{Tr}(A^*B)$.*

Proof. There are $3^2 = 9$ matrices of the form P^aQ^b, matching $\dim_{\mathbb{C}} M_3 = 9$; it therefore suffices to show that they are pairwise orthogonal.

Since $P^* = P^{-1}$ and $Q^* = Q^{-1}$, we have $(P^aQ^b)^* = Q^{-b}P^{-a}$, so

$$\langle P^aQ^b, P^cQ^d\rangle = \mathrm{Tr}\left(Q^{-b}P^{-a}P^cQ^d\right) = \mathrm{Tr}\left(Q^{-b}P^{c-a}Q^d\right).$$

Using the commutation relation $Q^kP^\ell = \omega^{-k\ell}P^\ell Q^k$, we obtain

$$Q^{-b}P^{c-a} = \omega^{b(c-a)}\, P^{c-a}Q^{-b},$$

and hence

$$\langle P^aQ^b, P^cQ^d\rangle = \omega^{b(c-a)}\,\mathrm{Tr}\left(P^{c-a}Q^{d-b}\right).$$

If $b \neq d$, then Q^{d-b} is a nontrivial permutation matrix, so $P^{c-a}Q^{d-b}$ has zero diagonal entries and $\mathrm{Tr}(P^{c-a}Q^{d-b}) = 0$.

If $b = d$, then

$$\langle P^aQ^b, P^cQ^b\rangle = \mathrm{Tr}(P^{c-a}).$$

Since $P^{c-a} = \mathrm{diag}(1, \omega^{c-a}, \omega^{2(c-a)})$,

$$\mathrm{Tr}(P^{c-a}) = 1 + \omega^{c-a} + \omega^{2(c-a)} = \begin{cases} 3, & a = c, \\ 0, & a \neq c, \end{cases}$$

using $1 + \omega + \omega^2 = 0$.

Combining both cases gives

$$\langle P^aQ^b, P^cQ^d\rangle = 3\,\delta_{ac}\delta_{bd},$$

so the nine matrices P^aQ^b are pairwise orthogonal, and therefore form an orthogonal basis of M_3. □

Excluding the identity matrix from this basis leaves eight traceless matrices, which together span $\mathfrak{sl}_3(\mathbb{C})$. A basis of $\mathfrak{su}_3$ is then obtained by passing to suitable skew-Hermitian linear combinations of these eight matrices, exactly as the passage from σ_i to $i\sigma_i$ produced a basis of $\mathfrak{su}(2)$ above. In this sense, the clock and shift matrices furnish a natural higher-dimensional generalisation of the Pauli matrices, suited to describing three-level quantum systems.

4.2 Classification of self-reflective subspaces generated by P and Q

By the results of the previous section, it suffices to focus on self-adjoint subspaces of $\mathfrak{sl}_3$. By Lemma 4.1.1, the matrices obtained as products of P and Q are pairwise orthogonal, so what remains is to classify the subspaces they span.

4.2.1 Fundamental reflection-invariant pairs

Since P and Q are unitary ($P^* = P^2$, $Q^* = Q^2$), the conjugate transpose of an arbitrary generator satisfies

$$(P^aQ^b)^* = \omega^{-ab}\,P^{3-a}Q^{3-b}.$$

Because any self-reflective subspace must be closed under conjugate transposition, each basis generator P^aQ^b must appear alongside its structural reflection partner

$P^{3-a}Q^{3-b}$. This partitions the eight non-identity generators into four mutually exclusive, reflection-invariant blocks of dimension two:

$$\begin{aligned}
\mathcal{P}_1 &= \{P, P^2\} && \text{(purely diagonal block)} \\
\mathcal{P}_2 &= \{Q, Q^2\} && \text{(purely shift block)} \\
\mathcal{P}_3 &= \{PQ, P^2Q^2\} && \text{(first mixed block)} \\
\mathcal{P}_4 &= \{P^2Q, PQ^2\} && \text{(second mixed block)}
\end{aligned}$$

For instance, the choice $S = \{1, 3\}$ gives the four-dimensional subspace $W_{1,3} = \operatorname{span}\{P, P^2, PQ, P^2Q^2\}$.

4.2.2 Action of the discrete Fourier transform

Let F denote the 3×3 discrete Fourier transform matrix. The unitary similarity transformation $X \mapsto FXF^*$ acts as an automorphism of the generalised Pauli algebra, exchanging clock and shift operators:

$$FPF^* = Q^2, \qquad FQF^* = P.$$

On the fundamental blocks $\mathcal{P}_k$, this induces a natural pair-swapping involution:

$$F\mathcal{P}_1F^* = \mathcal{P}_2, \qquad F\mathcal{P}_2F^* = \mathcal{P}_1, \qquad F\mathcal{P}_3F^* = \mathcal{P}_4, \qquad F\mathcal{P}_4F^* = \mathcal{P}_3.$$

4.2.3 Subspace classification

Because the fundamental building blocks $\mathcal{P}_k$ are non-overlapping 2-dimensional sets, every self-reflective subspace generated by powers of P and Q is formed by taking a direct sum of a subset of these blocks (including the empty set, which yields the zero subspace).

Proposition 4.2.1. *Every self-reflective subspace $V \subseteq M_3$ generated by P and Q is classified by its dimension $m = \dim(V)$, as recorded in Table 4.1.*

m	subspaces	count
$1, 3, 5, 7$	none exist	0
0	$V_0 = \{0\}$	1
2	$V_k = \operatorname{span}(\mathcal{P}_k)$, $k \in \{1, 2, 3, 4\}$	4
4	$W_{i,j} = \operatorname{span}(\mathcal{P}_i \oplus \mathcal{P}_j)$, $1 \le i < j \le 4$	6
6	$U_k = \operatorname{span}\left(\bigoplus_{i \neq k} \mathcal{P}_i\right)$, $k \in \{1, 2, 3, 4\}$	4
8	$\mathfrak{su}_3 = \operatorname{span}\left(\bigoplus_{k=1}^{4} \mathcal{P}_k\right)$	1

Figure 4.1: Classification of self-reflective subspaces $V \subseteq M_3$ generated by P and Q, by dimension.

Corollary 4.2.1. Including the trivial subspace V_0, there exist precisely 16 self-reflective subspaces generated by P and Q.

This classification reduces to elementary combinatorics once the blocks $\mathcal{P}_1, \dots, \mathcal{P}_4$ are fixed. Since these four blocks partition the eight non-identity Pauli generators into disjoint reflection-invariant pairs, choosing a self-reflective subspace V amounts to nothing more than choosing a subset $S \subseteq \{1, 2, 3, 4\}$ of blocks to include, with

$$V = \bigoplus_{k \in S} \mathcal{P}_k.$$

There are $2^4 = 16$ such subsets in total, matching the count in the corollary. Moreover, since each included block contributes exactly two dimensions, the resulting dimension $\dim(V)$ is always even, explaining why V can never have odd dimension. The counts in each row of the table are then simply the binomial coefficients $\binom{4}{r}$, for $r = |S|$ blocks chosen — namely $1, 4, 6, 4, 1$ for $r = 0, 1, 2, 3, 4$ — which is also why the table is symmetric under $m \leftrightarrow 8 - m$: choosing r blocks to include is exactly the same data as choosing $4 - r$ blocks to exclude.

4.2.4 Full unitary equivalence within each dimension

The action of the discrete Fourier transform F considered above only produces 4 orbits among the six four-dimensional subspaces $W_{i,j}$, namely $\{W_{1,2}\}, \{W_{3,4}\}, \{W_{1,3}, W_{2,4}\}, \{W_{1,4}, W_{2,3}\}$. We now show that this is an artifact of restricting attention to F alone: once a second natural unitary is brought in, *all six* subspaces of dimension 4 turn out to be unitarily equivalent – and, more generally, so are all subspaces of any fixed dimension $m \in \{0, 2, 4, 6, 8\}$.

Consider the diagonal phase matrix

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \omega \end{pmatrix}.$$

Since S is diagonal, it commutes with P, so $SPS^{-1} = P$ and hence S fixes the block $\mathcal{P}_1 = \{P, P^2\}$ point-wise. Its effect on Q is more interesting.

Lemma 4.2.1. *Conjugation by S cyclically permutes the blocks $\mathcal{P}_2, \mathcal{P}_3, \mathcal{P}_4$, while fixing $\mathcal{P}_1$ point-wise. Explicitly,*

$$SQS^{-1} = \omega^{-1} PQ, \qquad SQ^2S^{-1} = P^2Q^2,$$
$$S(PQ)S^{-1} = \omega^{-1} P^2 Q, \qquad S(P^2Q^2)S^{-1} = PQ^2.$$

Proof. Write e_1, e_2, e_3 for the standard basis, so that $Qe_i = e_{i+1}$ (indices mod 3) and $Se_i = s_i e_i$ with $s_1 = s_2 = 1$, $s_3 = \omega$. Then

$$(SQS^{-1})e_i = s_i^{-1}(SQ)e_i = s_i^{-1} s_{i+1} \, e_{i+1},$$

and computing the ratios s_{i+1}/s_i for $i = 1, 2, 3$ gives $1, \omega, \omega^2$ respectively. Comparing with $(PQ)e_i = p_{i+1}e_{i+1}$, where $p_1 = 1, p_2 = \omega, p_3 = \omega^2$, which gives coefficients $\omega, \omega^2, 1$, we see that SQS^{-1} has exactly the coefficients of PQ shifted by a global factor of ω^{-1}; that is, $SQS^{-1} = \omega^{-1}PQ$.

Squaring, and using the commutation relation $QP = \omega^{-1}PQ$ to simplify $(PQ)^2 = P(QP)Q = \omega^{-1}P^2Q^2$, we obtain

$$SQ^2S^{-1} = (SQS^{-1})^2 = \omega^{-2}(PQ)^2 = \omega^{-2} \cdot \omega^{-1}P^2Q^2 = P^2Q^2,$$

since $\omega^{-3} = 1$.

For the remaining two identities, use that S commutes with P:

$$S(PQ)S^{-1} = P\,(SQS^{-1}) = P \cdot \omega^{-1}PQ = \omega^{-1}P^2Q,$$

$$S(P^2Q^2)S^{-1} = P^2\,(SQ^2S^{-1}) = P^2 \cdot P^2Q^2 = P^4Q^2 = PQ^2,$$

using $P^3 = I$ in the last step. Collecting these four identities: S sends $\{Q, Q^2\} = \mathcal{P}_2$ onto $\{PQ, P^2Q^2\} = \mathcal{P}_3$ (up to phase), and sends $\mathcal{P}_3$ onto $\{P^2Q, PQ^2\} = \mathcal{P}_4$ (up to phase). By the same computation applied once more, or simply by counting (three blocks permuted among themselves by a single conjugation must be permuted cyclically or fixed point-wise, and we have already ruled out the latter), S also sends $\mathcal{P}_4$ onto $\mathcal{P}_2$. This proves the claim. □

In terms of the block indices $\{1, 2, 3, 4\}$, conjugation by F induces the permutation $\sigma_F = (1\,2)(3\,4)$ (as noted above), while conjugation by S induces the 3-cycle $\sigma_S = (2\,3\,4)$.

Remark 4.2.1. A basic fact from finite group theory will be used below: the alternating group A_4 (of order 12) has *no* subgroup of order 6. Indeed, a subgroup of order 6 would have index 2 in A_4 and hence be normal, but the only normal subgroups of A_4 are the trivial subgroup, the Klein four-group $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$, and A_4 itself – none of order 6.

Proposition 4.2.2. *The permutations σ_F and σ_S generate the full alternating group A_4 acting on the block indices $\{1, 2, 3, 4\}$. In particular this action is transitive on the six unordered pairs $\{i, j\} \subseteq \{1, 2, 3, 4\}$.*

Proof. Both σ_F (a double transposition) and σ_S (a 3-cycle) are even permutations, so the group H they generate is a subgroup of A_4. Since H contains an element of order 2 and an element of order 3, its order is divisible by 6. But A_4 has no subgroup of order 6 by Remark 4.2.1, and the only divisor of $|A_4| = 12$ that is a multiple of 6 and admits a subgroup is 12 itself; hence $H = A_4$.

It remains to check that A_4 is transitive on unordered pairs. The subgroup of A_4 fixing a point, say 1, is $\{e, (2\,3\,4), (2\,4\,3)\}$, which already acts transitively on the remaining three points $\{2, 3, 4\}$; hence A_4 is 2-transitive on $\{1, 2, 3, 4\}$ (transitive on ordered pairs, as $|A_4| = 12 = 4 \times 3$), and in particular transitive on unordered pairs. □

Corollary 4.2.2. For every dimension $m \in \{2, 4, 6\}$, any two of the subspaces of dimension m listed in Table 4.1 are related by conjugation $V = UWU^*$ for some $U \in U(3)$, built as a composition of the unitary matrices F and S (and their inverses). Together with the trivial cases $m = 0, 8$, this shows that all sixteen self-reflective subspaces generated by P and Q fall into exactly *five* classes under unitary conjugation, one for each dimension $m \in \{0, 2, 4, 6, 8\}$.

Proof. By Proposition 4.2.2, for any two pairs $\{i, j\}, \{i', j'\} \subseteq \{1, 2, 3, 4\}$ there is a permutation $\tau \in A_4$ with $\tau(\{i, j\}) = \{i', j'\}$, realized as a word in σ_F, σ_S and their inverses. Replacing each σ_F or σ_S in this word by the corresponding unitary F or S yields a unitary U with $U\mathcal{P}_iU^* = \mathcal{P}_{\tau(i)}$ for every block index i (up to scalar phases on individual generators, which do not affect the spans), and hence

$$UW_{i,j}U^* = U\big(\operatorname{span}(\mathcal{P}_i \oplus \mathcal{P}_j)\big)U^* = \operatorname{span}(\mathcal{P}_{\tau(i)} \oplus \mathcal{P}_{\tau(j)}) = W_{i',j'}.$$

The same argument, using only the transitivity of A_4 on single points, gives the case $m = 2$; the case $m = 6$ follows by taking orthogonal complements, since U_k is precisely the orthogonal complement of V_k inside $\mathfrak{su}_3$, and unitary conjugation commutes with taking orthogonal complements.

Finally, since U can always be rescaled by a scalar λ without changing the conjugation action $(\lambda U)(\cdot)(\lambda U)^* = U(\cdot)U^*$, one may choose λ so that $\lambda U \in SU(3)$, matching the form required by Gromada's theorem (Theorem 3.5.1) for the corresponding quantum graphs to be isomorphic. □

4.3 Classification of loop-free and undirected graphs on M_3

In this section we exhibit, for each possible number of quantum edges $m \in \{0, 2, 4, 6, 8\}$, a unique quantum graph that can be represented as a derived quantum graph of a voltage quantum graph. Thanks to a result by Schäfer and Tobolski [9], namely Theorem 3.4.1, those graphs are quantum isomorphic to quantum graphs on $(\mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n, \psi_{\rtimes})$, which, in turn, is isomorphic to $(M_3, 3\,\mathrm{Tr})$ by Lemma 3.1.2, and so can be thought of as graphs on $(M_3, 3\,\mathrm{Tr})$. A result by Schäfer and Tobolski [9] tells us that all those graphs are quantum isomorphic to classical graphs on 9 vertices.

Proposition 4.3.1. *There are four nontrivial, loop-free and undirected graphs on $(M_3, 3\,\mathrm{Tr})$ up to isomorphism. Their matrices, expressed in the basis of generalised and normalised Pauli matrices (Subsection 4.1.2) using the formula from Theorem 3.5.1, are as follows:*

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

Matrix of the graph obtained from $V_2(m = 2)$

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & \omega^2 & \omega & 0 & 0 \\ 0 & 0 & -1 & \omega & 0 & 0 & 0 & \omega^2 & 0 \\ 0 & 0 & \omega^2 & -1 & 0 & 0 & 0 & \omega & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & \omega & 0 & 0 & 0 & -1 & \omega^2 & 0 & 0 \\ 0 & \omega^2 & 0 & 0 & 0 & \omega & -1 & 0 & 0 \\ 0 & 0 & \omega & \omega^2 & 0 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix}$$

Matrix of the graph obtained from $W_{24}(m = 4)$

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix}$$

Matrix of the graph obtained from $U_1(m = 6)$

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 2 \end{pmatrix}$$

Matrix of the graph obtained from $\mathfrak{sl}_3(m = 8)$

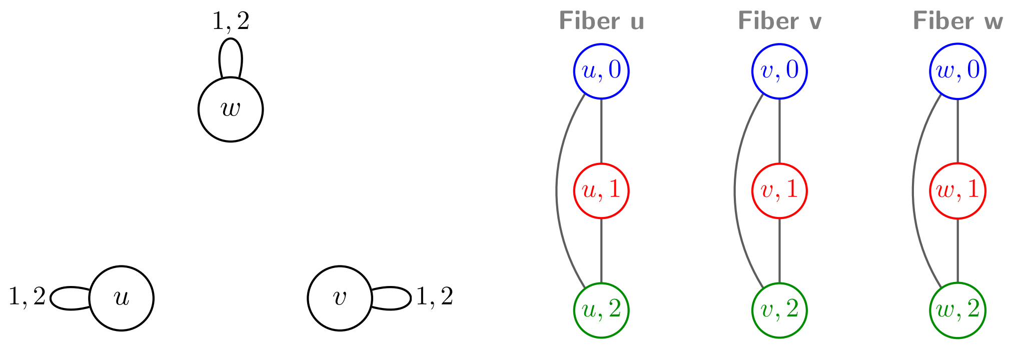
Figure 4.2: Adjacency matrices, expressed in the generalised Pauli basis via Theorem 3.5.1, of the four nontrivial loop-free and undirected quantum graphs on $(M_3, 3\,\mathrm{Tr})$ up to isomorphism, one for each nontrivial dimension $m \in \{2, 4, 6, 8\}$.

The trivial case $m = 0$, corresponding to the empty graph, is omitted, since it carries no meaningful structure to discuss.

The matrices in Proposition 4.3.1 were obtained by rote computation. For each possible subspace as described in Table 4.1 we calculated the appropriate matrix using the formula given in Theorem 3.5.1 and compared them with matrices describing derived quantum graphs coming from quantum voltage graphs $\tilde{A} = \{A_k\}_{k \in \hat{\mathbb{Z}}_3}$ given by the formula

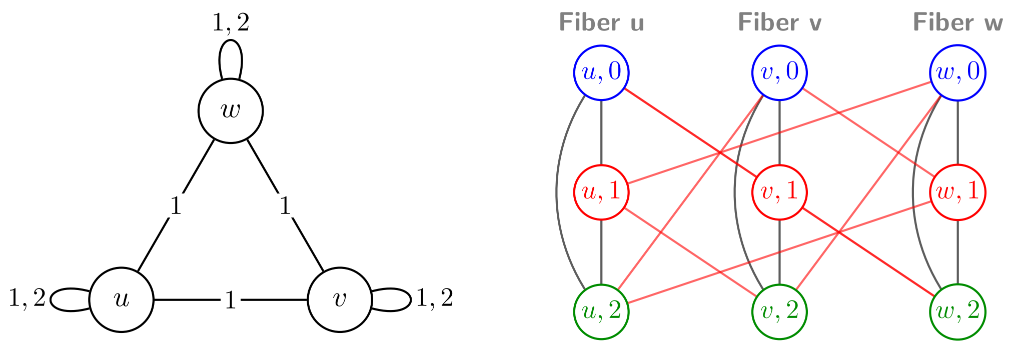
$$\tilde{A} \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_3 := \sum_{k \in \hat{\mathbb{Z}}_3} m\big(X_k \otimes E_{\rtimes}^* \tilde{A}_k E_{\rtimes}\big)\, m^*.$$

The following figures depict the voltage graphs and derived graphs that are quantum isomorphic to the graphs given as matrices in Figure 4.2. Concretely, the voltage graph with only self-loops (labeled $1, 2$ at every vertex) corresponds to the case $m = 2$; adding a single 1-voltage triangle to these loops gives $m = 4$; adding both a 1- and a 2-voltage triangle gives $m = 6$; and the full voltage graph, with edges of every voltage $0, 1, 2$ between every pair of vertices, gives $m = 8$, the complete quantum graph $\mathfrak{sl}_3$.



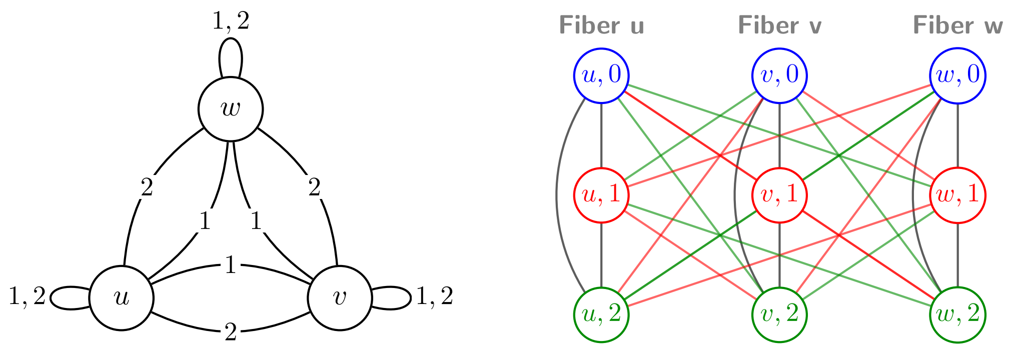
Base Γ: undirected loops Derived Γ^λ: undirected 3-cycles

Figure 4.3: Voltage graph and derived cover for the loopfree, undirected quantum graph on M_3 with $m = 2$ quantum edges.



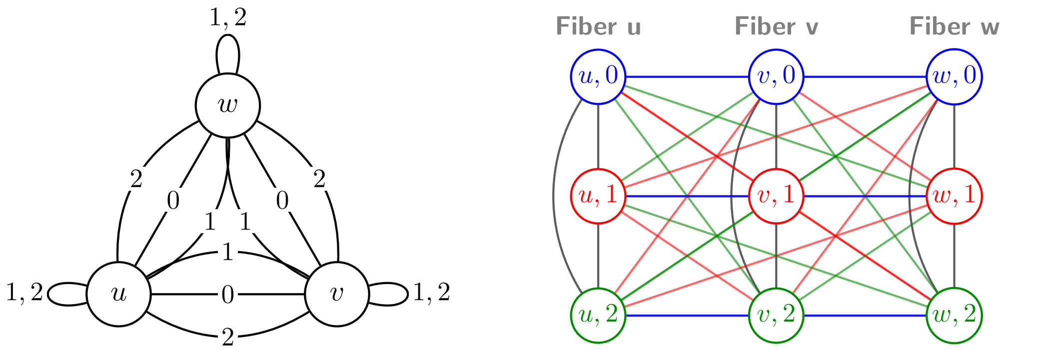
Base Γ: undirected triangle & loops Derived Γ^λ: exactly 18 undirected edges

Figure 4.4: Voltage graph and derived cover for the loopfree, undirected quantum graph on M_3 with $m = 4$ quantum edges.



Base Γ: undirected multigraph Derived Γ^λ: partial $K_{3,3}$ between fibers

Figure 4.5: Voltage graph and derived cover for the loopfree, undirected quantum graph on M_3 with $m = 6$ quantum edges.



Base Γ: full undirected multigraph   Derived Γ^λ: perfect $K_{3,3}$ between all fibers

Figure 4.6: Voltage graph and derived cover for the loopfree, undirected quantum graph on M_3 with $m = 8$ quantum edges.

Since each of these voltage quantum graphs arises, via Theorem 3.4.1, as a trivial-action derived quantum graph of an ordinary (classical) voltage graph, the discussion above in fact exhibits an explicit quantum isomorphism between each nontrivial loop-free, undirected quantum graph on $(M_3, 3\,\mathrm{Tr})$ and a classical derived graph on the nine vertices $\{u, v, w\} \times \mathbb{Z}_3$ shown on the right of each figure – concretely realizing the abstract existence statement from the start of this section. Combined with Corollary 4.2.2, this completes the classification of loop-free, undirected quantum graphs on $(M_3, 3\,\mathrm{Tr})$ up to quantum isomorphism: every such graph is quantum isomorphic to exactly one of the five classical graphs exhibited in this section (including the trivial $m = 0$ case).

4.4 Vertex-transitive graphs

Classically, a graph is called vertex-transitive if, for any two vertices v, w, there exists a graph automorphism f with $f(v) = w$. This notion was later extended to the quantum setting, and in their recent paper Hayes et al. [6] classify all vertex-transitive quantum graphs on M_3 up to isomorphism.

Definition 4.4.1 (Vertex-transitive graphs, Hayes et al. [6], Definition 3.3)**.** A quantum graph A is called *vertex-transitive* if the set

$$\{u \in U(n) : uVu^* = V\}$$

spans M_n, where $V \subseteq M_n$ is the subspace associated with A.

The graphs obtained in Proposition 4.3.1 are all vertex-transitive, which is proven in the next Proposition 4.4.1.

Proposition 4.4.1. *Every self-reflective subspace $V \subseteq M_3$ generated by P and Q (i.e. every direct sum of a subset of the blocks $\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3, \mathcal{P}_4$) gives rise to a vertex-transitive quantum graph.*

Proof. We first show that conjugation by any element P^aQ^b, $a, b \in \{0, 1, 2\}$, preserves each one-dimensional span$\{P^cQ^d\}$ individually. Using the commutation relation $PQ = \omega\, QP$, one computes

$$Q^bPQ^{-b} = \omega^{-b}P, \qquad P^aQP^{-a} = \omega^aQ,$$

and more generally, for any generator P^cQ^d,

$$(P^aQ^b)\, P^cQ^d\, (P^aQ^b)^{-1} = \omega^{\,bc-ad}\, P^cQ^d.$$

Thus conjugation by P^aQ^b fixes the generator P^cQ^d up to a scalar phase ω^{bc-ad}, and in particular sends span$\{P^cQ^d\}$ to itself.

Since every block $\mathcal{P}_k$ is a set of two such generators, and every self-reflective subspace V is a direct sum of some subset of the blocks, it follows that

$$(P^aQ^b)\, V\, (P^aQ^b)^{-1} = V$$

for *every* $a, b \in \{0, 1, 2\}$, i.e. the full group $\{P^aQ^b\}_{a,b=0,1,2}$ stabilizes V, regardless of which subset of blocks was chosen.

But we have already shown that the nine matrices P^aQ^b form an orthogonal basis of M_3 (Lemma 4.1.1); in particular they span M_3. Since

$$\{P^aQ^b\}_{a,b=0,1,2} \subseteq \{u \in U(3) : uVu^* = V\},$$

the stabilizer of V already spans M_3 on its own, so V is vertex-transitive. □

Remark 4.4.1. In particular, this covers all sixteen subspaces of Table 4.1, including the extremal cases $V_0 = \{0\}$ and $\mathfrak{su}_3$, whose stabilizers are trivially all of $U(3)$. More strikingly, the proposition shows that vertex-transitivity holds already at the *minimal* nontrivial dimension $m = 2$: even the smallest quantum graphs $V_k =$ span$(\mathcal{P}_k)$ are vertex-transitive, stabilized by the same 9-element group that stabilizes every other subspace in the classification.

The following Table shows the correspondences between the quantum graphs in Proposition 4.3.1 as well as the zero graph and the graphs from Hayes et al. [6, Figure 1].

m	Our	Theirs
$m = 0$	the trivial quantum graph	VT_0^3
$m = 2$		VT_2^3
$m = 4$		VT_4^3
$m = 6$		VT_6^3
$m = 8$		VT_8^3

Figure 4.7: Our graphs vs their graphs.

4.5 Classification of different types of graphs

In this section, we examine the prospects for classifying quantum graphs without assuming them to be loop-free or undirected. Theorem 4.5.1 demonstrates that simple quantum graphs on M_2 can be fully classified, establishing that there are only finitely many such graphs up to isomorphism.

Theorem 4.5.1 (Gromada [3], Theorem 3.11)**.** *A simple graph on M_2 is determined up to isomorphism solely by the number of quantum edges $m \in \{0, 1, 2, 3\}$.*

Proof. Recall first that $V_0 = M_2(\mathbb{C})$ is assumed to carry no loops, which implies that a simple quantum graph on $(M_2(\mathbb{C}), \psi)$ admits at most $2^2 - 1 = 3$ quantum edges.

By Theorem 3.5.1, any simple quantum graph on $M_2(\mathbb{C})$ with m quantum edges is uniquely determined by an m-dimensional self-adjoint subspace $V \subset \mathfrak{sl}_2(\mathbb{C})$. By Lemma 3.5.2, it suffices to regard V as a real vector space, or equivalently, to identify iV with a subspace of the Lie algebra $\mathfrak{su}(2)$; passing to the standard basis given by the Pauli matrices, we may then naturally view V as a real subspace $V \subset \mathbb{R}^3$.

By Theorem 3.5.1, two such subspaces $V, W \subset \mathbb{R}^3$ yield isomorphic quantum graphs if and only if they are related by conjugation by a unitary matrix $U \in SU(2)$. Under the identification of $\mathfrak{su}(2)$ with $\mathbb{R}^3$, this conjugation action of $SU(2)$ corresponds precisely to standard rotations, via the canonical isomorphism $PU(2) \cong SO(3)$.

Consequently, the quantum graphs corresponding to V and W are isomorphic if and only if V can be mapped onto W by a rotation in $SO(3)$. Since two real subspaces of $\mathbb{R}^3$ are congruent under $SO(3)$ if and only if they have the same dimension, the isomorphism class of the quantum graph is completely determined by $m = \dim V$. □

The proof of this theorem relies on the well-known isomorphism $PU(2) \cong SO(3)$, which plays a key role throughout. This result does not generalise to higher dimensions, and in particular to dimension 3, as the following lemma shows.

Lemma 4.5.1. *The natural action*

$$PU(3) \curvearrowright \mathbb{P}(\mathbb{R}^8) = \{W \subseteq \mathbb{R}^8 : \dim(W) = 1\}$$

has infinitely many orbits.

Proof. Let $V = \mathfrak{su}(3) \cong \mathbb{R}^8$, and identify $\mathbb{P}(\mathbb{R}^8)$ with $\mathbb{P}(V)$. The action of $PU(3)$ on V is the adjoint representation

$$g \cdot X = gXg^{-1},$$

which induces an action on the 1-dimensional subspaces of V. To show that this action has infinitely many orbits, it suffices to exhibit a projective invariant taking infinitely many values.

Define

$$q(X) = -\operatorname{Tr}(X^2), \qquad c(X) = i \operatorname{Tr}(X^3).$$

Since the trace is invariant under conjugation,

$$q(gXg^{-1}) = q(X), \qquad c(gXg^{-1}) = c(X)$$

for every $g \in PU(3)$. Moreover, for every $\lambda \in \mathbb{R}$,

$$q(\lambda X) = \lambda^2 q(X), \qquad c(\lambda X) = \lambda^3 c(X),$$

so the ratio

$$I(X) = \frac{c(X)^2}{q(X)^3}$$

is invariant both under the adjoint action and under rescaling by nonzero scalars, and hence defines a well-defined invariant of the induced action on $\mathbb{P}(V)$.

Consider now the family

$$X_a = i\operatorname{diag}(a, 1, -a-1), \qquad a \in \mathbb{R}.$$

A direct computation gives

$$q(X_a) = 2(a^2 + a + 1), \qquad c(X_a) = -3a(a+1),$$

and therefore

$$I(X_a) = \frac{9a^2(a+1)^2}{8(a^2+a+1)^3}.$$

This is a non-constant continuous function of a; for instance,

$$I(X_0) = 0, \qquad I(X_1) = \frac{1}{6},$$

so $I(X_a)$ takes infinitely many values as a ranges over $\mathbb{R}$. Since I is constant on each $PU(3)$-orbit of $\mathbb{P}(V)$, distinct values of I must correspond to distinct orbits. Hence the action of $PU(3)$ on $\mathbb{P}(\mathbb{R}^8)$ has infinitely many orbits. □

Thus the argument used in the proof of Theorem 4.5.1 cannot be used. There is a much wider variety of graphs on M_3 than on M_2.

Proposition 4.5.1 (Gromada [3], Proposition 3.14)**.** *For any $m = 1, 2, 3$, there is an infinite family of mutually non-isomorphic undirected quantum graphs containing m quantum edges. They can be chosen as follows:*

- *For $m = 1$, we define $A(t) = P_3(t)$, $t \in [0, \pi/2]$.*
- *For $m = 2$, we define $A(t) = P_3(t) + P_2$, $t \in [0, \pi/2]$.*

- *For $m = 3$, we define $A(t) = P_3(t) + P_2 + P_1$, $t \in [0, \pi/2]$.*

Here, we denote $P_i = \sigma_i \otimes \sigma_i^*$, where σ_i are the Pauli matrices of Example 2.3.2. In addition, we define $P_3(t) = \sigma_3(t) \otimes \sigma_3(t)^*$, where $\sigma_3(t) = \sigma_3 \sin t + I_2 \cos t$.

Every other undirected quantum graph is isomorphic to one of these families (or is equal to the empty graph or the full graph).

To be more concrete, we can write down explicitly

$$P_3(t) = \begin{pmatrix} 1+\sin(2t) & 0 & 0 & 0 \\ 0 & \cos(2t) & 0 & 0 \\ 0 & 0 & \cos(2t) & 0 \\ 0 & 0 & 0 & 1-\sin(2t) \end{pmatrix}.$$

For $t = 0$, we obtain I_2, representing the pure loop. For $t = \pi/2$, we obtain P_3, corresponding to a loopless edge. For intermediate values $t \in (0, \pi/2)$, we obtain a family interpolating between these two configurations.

Proof. By Theorem 3.5.1, quantum graphs on $M_2(\mathbb{C})$ with m quantum edges stand in one-to-one correspondence with m-dimensional subspaces $V \subseteq \mathfrak{gl}_2(\mathbb{C})$. Since the quantum graphs are assumed to be undirected, Theorem 3.5.1 implies that V is self-adjoint ($V = V^*$).

Using the decomposition $\mathfrak{gl}_2(\mathbb{C}) = \mathbb{C} \cdot I_2 \oplus \mathfrak{sl}_2(\mathbb{C})$, any self-adjoint subspace V admits an orthonormal basis of elements of the form

$$v = \lambda I_2 + w, \qquad \lambda \in \mathbb{R}, \quad w \in \mathfrak{su}(2).$$

Identifying $\mathfrak{su}(2)$ with $\mathbb{R}^3$ via the Pauli basis $\{\sigma_1, \sigma_2, \sigma_3\}$, each matrix $\sigma_3(t) = \sigma_3 \sin t + I_2 \cos t$ corresponds to a vector in $\mathbb{R}^4 \cong \mathbb{R} \cdot I_2 \oplus \mathbb{R}^3$ with scalar component $\cos t$ and vector component $\sin t \cdot e_3$.

According to Theorem 3.5.1, two subspaces V, W yield isomorphic quantum graphs if and only if $V = UWU^*$ for some unitary $U \in SU(2)$. Under this adjoint action, U acts trivially on the scalar subspace $\mathbb{R} \cdot I_2$ and acts by standard rotations via $PU(2) \cong SO(3)$ on the vector space $\mathbb{R}^3$.

Because rotations in $SO(3)$ preserve the norm of the vector component $|\sin t|$ as well as the scalar component $\cos t$, two elements $\sigma_3(t_1)$ and $\sigma_3(t_2)$ with $t_1, t_2 \in [0, \pi/2]$ lie in the same $SU(2)$-orbit if and only if $t_1 = t_2$. Extending this by adding the fixed orthogonal direction matrices $P_2 = \sigma_2 \otimes \sigma_2^*$ and $P_1 = \sigma_1 \otimes \sigma_1^*$, the subspaces

$$V(t) = \mathrm{span}_{\mathbb{C}}\{\sigma_3(t), \sigma_2, \sigma_1\}_{\text{first } m \text{ elements}}$$

are mutually non-conjugated under $SU(2)$ for distinct values of $t \in [0, \pi/2]$. Hence, $A(t) = \sum_{s=1}^{m} \xi_s(t) \otimes \xi_s(t)^*$ defines a continuous, infinite family of mutually non-isomorphic quantum graphs for each $m \in \{1, 2, 3\}$.

Finally, given any m-dimensional self-adjoint subspace $V \subseteq \mathfrak{gl}_2(\mathbb{C})$, we can apply an appropriate rotation $U \in SU(2)$ to align its non-scalar projections with the standard Pauli basis vectors. Up to swapping generators and scalar shifts, every such subspace is conjugated under $SU(2)$ to precisely one subspace in these parameterized families (or to the trivial subspaces corresponding to the empty or full graph). $\square$

4.5.1 Simple quantum graph that is not quantum isomorphic to a classical graph

Most of the quantum graphs we have been dealing with have been quantum isomorphic to classical graphs. We will explicitly show a quantum graph on M_3 that is not isomorphic to any classical graph.

Example 4.5.1 (Gromada [3], Example 8.5). The matrix

$$\lambda_8 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix},$$

which is one of the *Gell–Mann matrices* spanning the Lie algebra $\mathfrak{su}_3$. Then the adjacency matrix $A = \lambda_8 \otimes \lambda_8$ defines a simple graph on M_3.

Chapter 5

Conclusion

blabla

5.1 Further research

one could try to classify graphs on M_n but it looks kinda tough ngl

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